

syzkaller / sanitizers: status update

Dmitry Vyukov (dvyukov@)
Linux Plumbers, Aug 26 2020

Agenda

- sanitizers
- syzkaller/syzbot
- stats and graphs

KCSAN: ConcurrencySanitizer

Detects **data races**.

Upstream since 5.8.

See [Data-race detection in the Linux kernel](#) talk.

KMSAN: MemorySantizer

Detects **uses of uninitialized memory**.

Not upstream (github.com/google/kmsan)

Rebased, deployed on syzbot.

GWP-ASAN -> KFENCE

Low-overhead sampling memory error detector (see [LPC2019 talk](#)).

RFC "soon".

KASAN: AddressSanitizer

`CONFIG_KASAN_VMALLOC=y`

`CONFIG_VMAP_STACK=y`

KASAN: AddressSanitizer

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BUG: KASAN: **use-after-free** in afs_manage_cell

...

Freed by task 3903:

kfree	mm/slab.c:3756
rcu_do_batch	kernel/rcu/tree.c:2428
rcu_core	kernel/rcu/tree.c:2656
__do_softirq	kernel/softirq.c:298

KASAN: AddressSantizer

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Last call_rcu() :

call_rcu	kernel/rcu/tree.c:2968
afs_manage_cell	fs/afs/cell.c:751
process_one_work	kernel/workqueue.c:2269
worker_thread	kernel/workqueue.c:2415

KASAN: AddressSanitizer

[2f004eea0fc8](#) x86/kasan: Print original address on #GP

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general protection fault for non-canonical address **0xdffffc0000000000**

general protection fault for non-canonical address **0xdffffc000000000c**

general protection fault for non-canonical address **0xdffffc00e0dffffe**

KASAN: AddressSantizer

[2f004eea0fc8](#) x86/kasan: Print original address on #GP

```
general protection fault for non-canonical address 0xdffffc0000000000  
KASAN: null-ptr-deref in range [0x000000000000-0x000000000007]
```

```
general protection fault for non-canonical address 0xdffffc000000000c  
KASAN: null-ptr-deref in range [0x0000000000060-0x0000000000067]
```

```
general protection fault for non-canonical address 0xdffffc00e0dffffe  
KASAN: user-memory-access in range [0x000706fffff0-0x000706fffff7]
```

KASAN+ARM MTE

[\[PATCH 00/35\]](#) kasan: add hardware tag-based mode for arm64

[MTE \(Memory Tagging Extensions\)](#)

syzkaller + syzbot

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syzkaller - OS kernel fuzzer:

- code-coverage-guided
- input-structure-aware
- focus on automation
- multi-OS

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syzbot - syzkaller automation:

- continuous kernel/syzkaller update
- bug reporting / tracking
- web dashboard

syzkaller.appspot.com

New ARCHes

- amd64
- 386
- arm
- arm64
- ppc64le

contributed by:
Alexander Egorenkov
Tobias Klauser
Jouni Hogander

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- **riscv64**
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 - **s390x**
- } not tested on syzbot

Moar kernel interface descriptions

- `watch_queue`, `copy_file_range`, `process_madvise`, `pidfd_getfd`, `preadv2`
- `/dev/snd/hw*`, `/dev/sequencer`, `/dev/raw`
- `wireguard`
- `exfat`, `afs`
- `v4l2: vim2m/vim2m2`
- `CAN/j1939`
- `netlabel/contrack/ipset`
- `lwtunnel_encap`
- `NFNL_SUBSYS_ (NFTABLES|CTNETLINK|ACCT|OCF|ULOG|QUEUE|CTHELPER)`
- `RTM_ (NEW|DEL|GET) (MDB|VLAN|ADDRLABEL|LINKPROP)`
- `NETLINK_ (RDMA|AUDIT|SOCK_DIAG)`
- `IPPROTO_ (MPTCP|L2TP)`
- `AF_PHONET`
- `wirt_wifi`, `vlan`, `macvlan`, `ipvlan`, `xfrm`, `vlan`, `macvlan`, `ipvlan`, `mactap`, `geneve`, `macvtap`, `batadv`
- `more BPF`
- ...

Bluetooth

- inject external packets via `/dev/vhci`

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- pre-setup VHCI in test process
 - open(/dev/hvci)
 - socket(AF_BLUETOOTH, SOCK_RAW, BTPROTO_HCI)
 - ioctl's + read's + write's + thread's

BlueTooth

- inject external packets via /dev/vhci
- pre-setup VHCI in test process
 - open(/dev/hvci)
 - socket(AF_BLUETOOTH, SOCK_RAW, BTPROTO_HCI)
 - ioctl's + read's + write's + thread's
- increases coverage of other BlueTooth parts

BlueTooth Bugs (<1 month)

KASAN: use-after-free Write in __sco_sock_close
KASAN: use-after-free Write in sco_chan_del
KASAN: use-after-free Read in hci_chan_del
KASAN: use-after-free Read in hci_send_acl
KASAN: use-after-free Read in __sco_sock_close
KASAN: use-after-free Read in hci_get_auth_info
KASAN: use-after-free Write in hci_conn_del
KASAN: use-after-free Read in __queue_work
KASAN: slab-out-of-bounds Read in lock_sock_nested
KASAN: slab-out-of-bounds Read in hci_inquiry_re...
KASAN: slab-out-of-bounds Read in hci_extended_i...
KASAN: slab-out-of-bounds Read in hci_le_meta_evt
KASAN: null-ptr-deref in amp_read_loc_assoc_final..
KASAN: null-ptr-deref Write in l2cap_chan_put
BUG: corrupted list in kobject_add_internal
BUG: unable to handle paging request in lock_sock_
BUG: corrupted list in hci_chan_del
BUG: corrupted list in bt_accept_unlink
memory leak in hci_conn_add
memory leak in read_adv_mon_features

KMSAN: uninit-value in process_adv_report
KMSAN: uninit-value in hci_chan_lookup_handle
general protection fault in hci_phy_link_complete_
general protection fault in hci_event_packet
general protection fault in bt_accept_unlink
WARNING: refcount bug in l2cap_global_chan_by_psm
WARNING: ODEBUG bug in put_device
WARNING in cancel_delayed_work
WARNING: ODEBUG bug in cancel_delayed_work
WARNING: ODEBUG bug in bt_host_release
WARNING: locking bug in hci_dev_reset
WARNING in hci_conn_timeout
WARNING: locking bug in l2cap_chan_del
WARNING: refcount bug in bt_accept_dequeue
INFO: register non-static key in l2cap_chan_del
INFO: register non-static key in l2cap_chan_close
INFO: register non-static key in skb_dequeue
INFO: register non-static key in skb_queue_purge
inconsistent lock state in sco_conn_del
inconsistent lock state in sco_sock_timeout

io_uring

```
fd = io_uring_create(sq_size, cq_size, ...)  
addr = mmap(..., [specific_size], ..., fd, ...)  
addr[specific_offset1] = specific_value1  
addr[specific_offset2] = specific_value3  
addr[specific_offset3] = specific_value3
```

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```

Pseudo-syscalls to the rescue!

```
# Submit sqe into the sq_ring  
sys_io_uring_submit(  
    ring_ptr ring_ptr,  
    sques_ptr sques_ptr,  
    sqe ptr[in, io_uring_sqe],  
    sques_index int32)
```

io_uring

KASAN: use-after-free Read in io_uring_setup (2)
KASAN: use-after-free Read in io_async_task_func
BUG: NULL pointer dereference in loop_rw_iter
general protection fault in io_poll_double_wake
possible deadlock in io_timeout_fn
possible deadlock in __io_queue_deferred
possible deadlock in io_queue_linked_timeout
KCSAN: data-race in __io_cqring_fill_event/io_uring_poll
INFO: task can't die in io_uring_flush
memory leak in io_submit_sqes
memory leak in rw_copy_check_uvector

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BUG: kernel NULL pointer dereference

Call Trace:

loop_rw_iter fs/io_uring.c:2829
io_write+0x6a2/0x7a0 fs/io_uring.c:3190
io_issue_sqe+0x1b0/0x60d0 fs/io_uring.c:5530
io_wq_submit_work+0x183/0x3d0 fs/io_uring.c:5775
io_worker_handle_work+0xa45/0x13f0 fs/io-wq.c:527
io_wqe_worker+0xbf0/0x10e0 fs/io-wq.c:569
kthread+0x3b5/0x4a0 kernel/kthread.c:292

Fix Bisection

If we suspect a bug is fixed -> find what fixed it!

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```
Subject: Re: WARNING: locking bug in try\_to\_grab\_pending  
Date: Fri, 14 Aug 2020 06:17:07 -0700  
Message-ID: <000000000000db6ee05acd63ca2@google.com> (raw)  
In-Reply-To: <0000000000006dc0290581ca413e@google.com>
```

syzbot suspects this issue was fixed by commit:

```
commit 1378817486d6860f6a927f573491afe65287abf1  
Author: Eric Dumazet <edumazet@google.com>  
Date: Thu May 21 18:29:58 2020 +0000
```

tipc: block BH before using dst_cache

```
bisection log: https://syzkaller.appspot.com/x/bisect.txt?x=175599f6900000  
start commit: 6663cf82 flow_offload: Fix flow action infrastructure  
git tree: net-next  
kernel config: https://syzkaller.appspot.com/x/.config?x=8572a6e4661225f4  
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syz repro: https://syzkaller.appspot.com/x/repro.syz?x=13e932a8c00000
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If the result looks correct, please mark the issue as fixed by replying with:

```
#syz fix: tipc: block BH before using dst_cache
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mailed 80 results

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Config Bisection

contributed by:
Jukka Kaartinen
Jouni Hogander

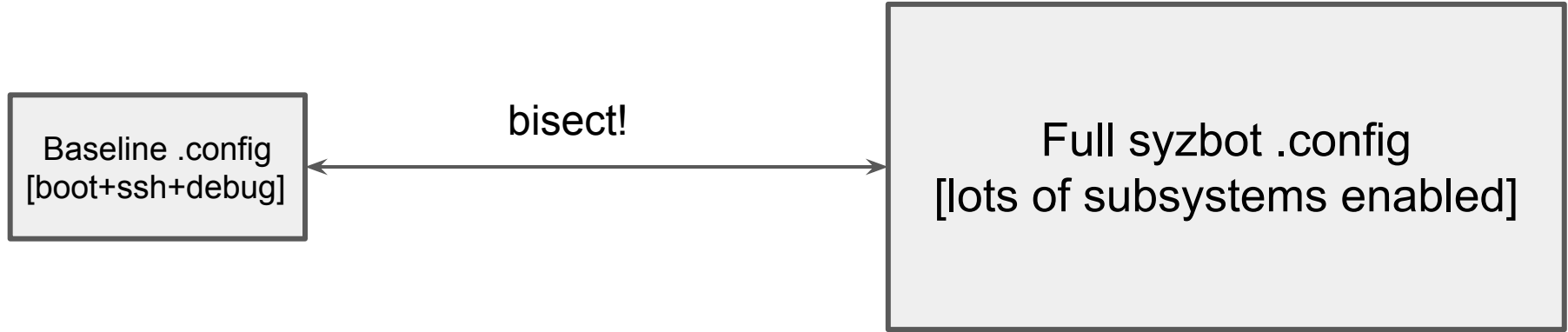
Full syzbot .config
[lots of subsystems enabled]

Config Bisection

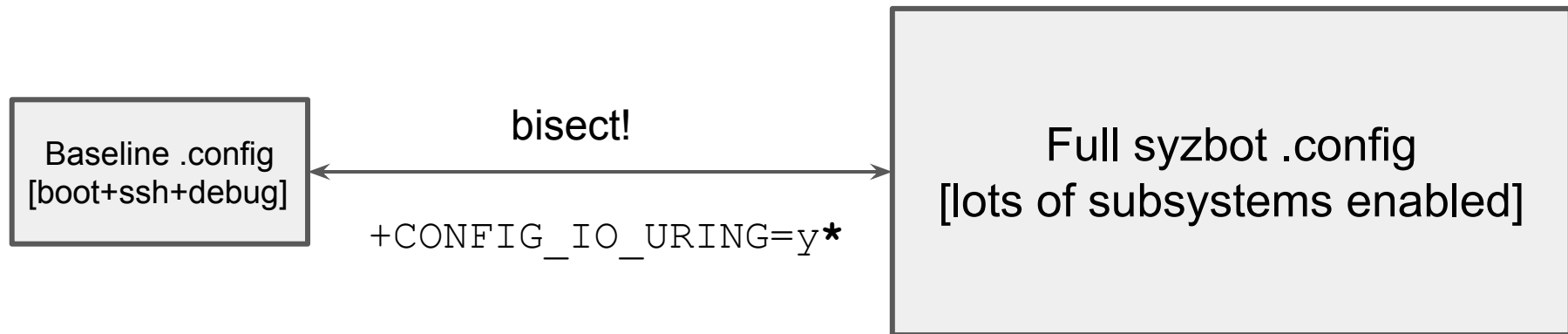
Baseline .config
[boot+ssh+debug]

Full syzbot .config
[lots of subsystems enabled]

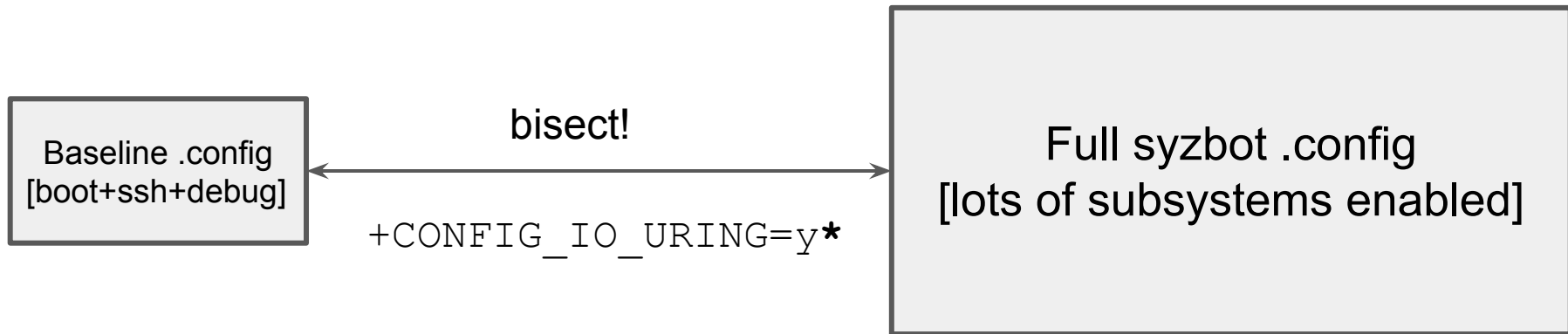
Config Bisection



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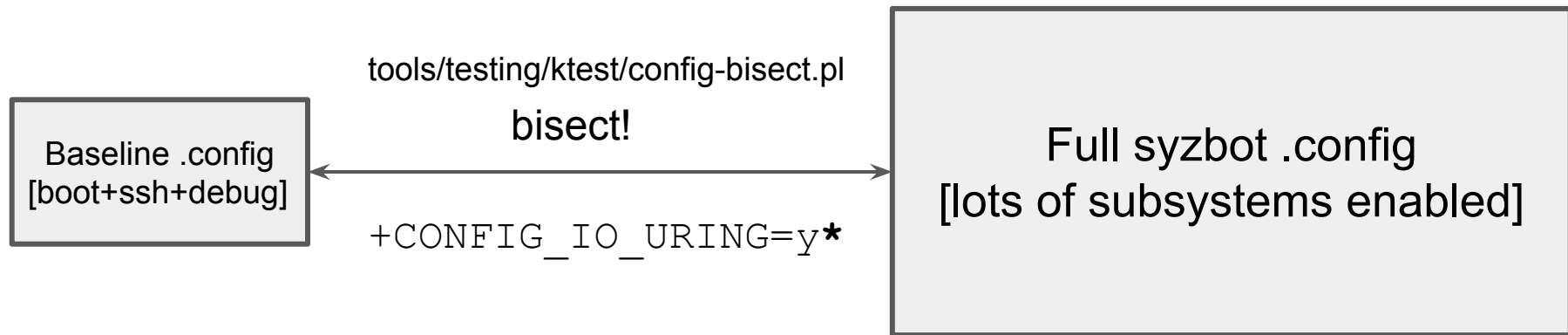


Config Bisection



- + faster builds
- + fewer unrelated crashes
- + fewer broken builds/boots
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More Automation

- Remind about unexisting fixing commit
 - can't be discovered anywhere for 90 days

More Automation

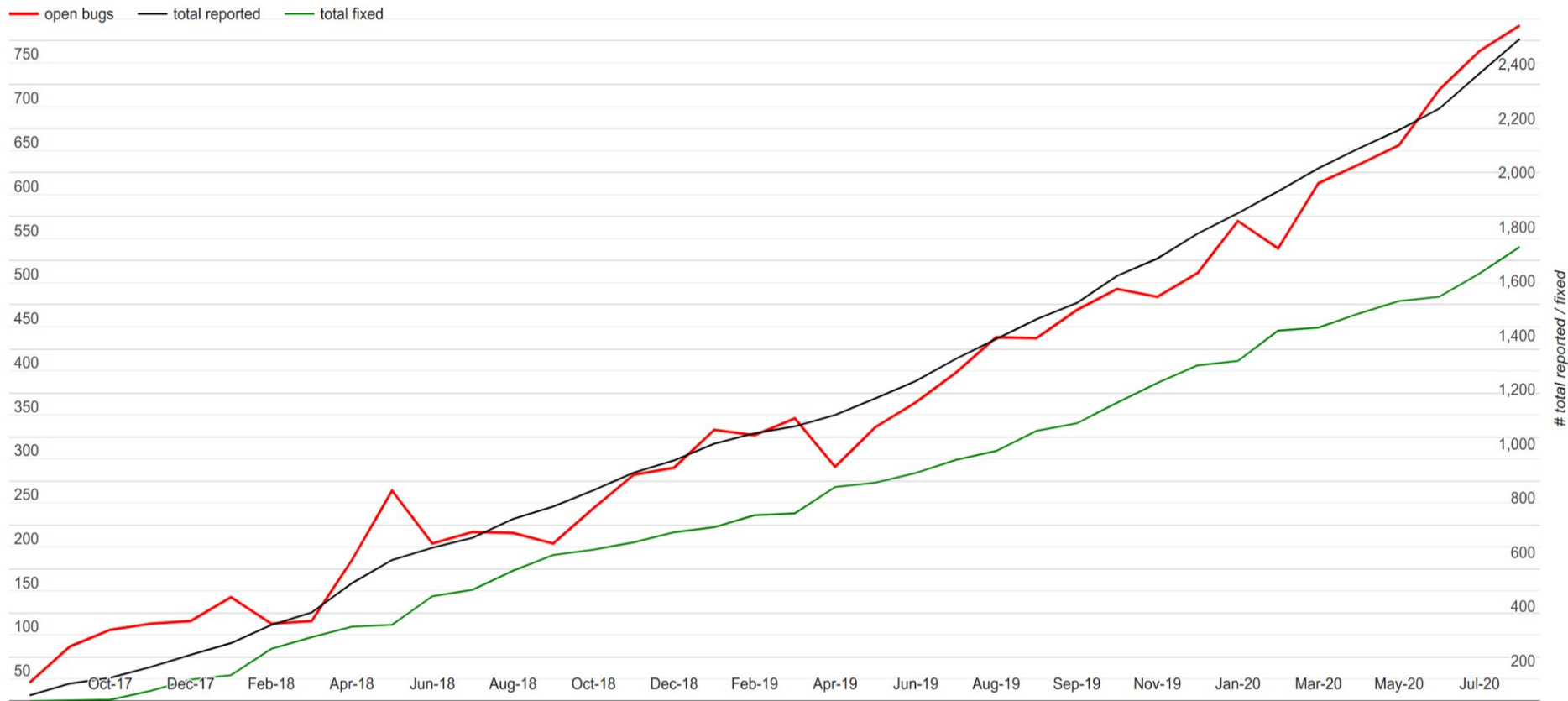
- Remind about unexisting fixing commit
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- Auto-obsoleting
 - no reproducer
 - upstream: 80-120 days; linux-next: 40-60 days
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More Automation

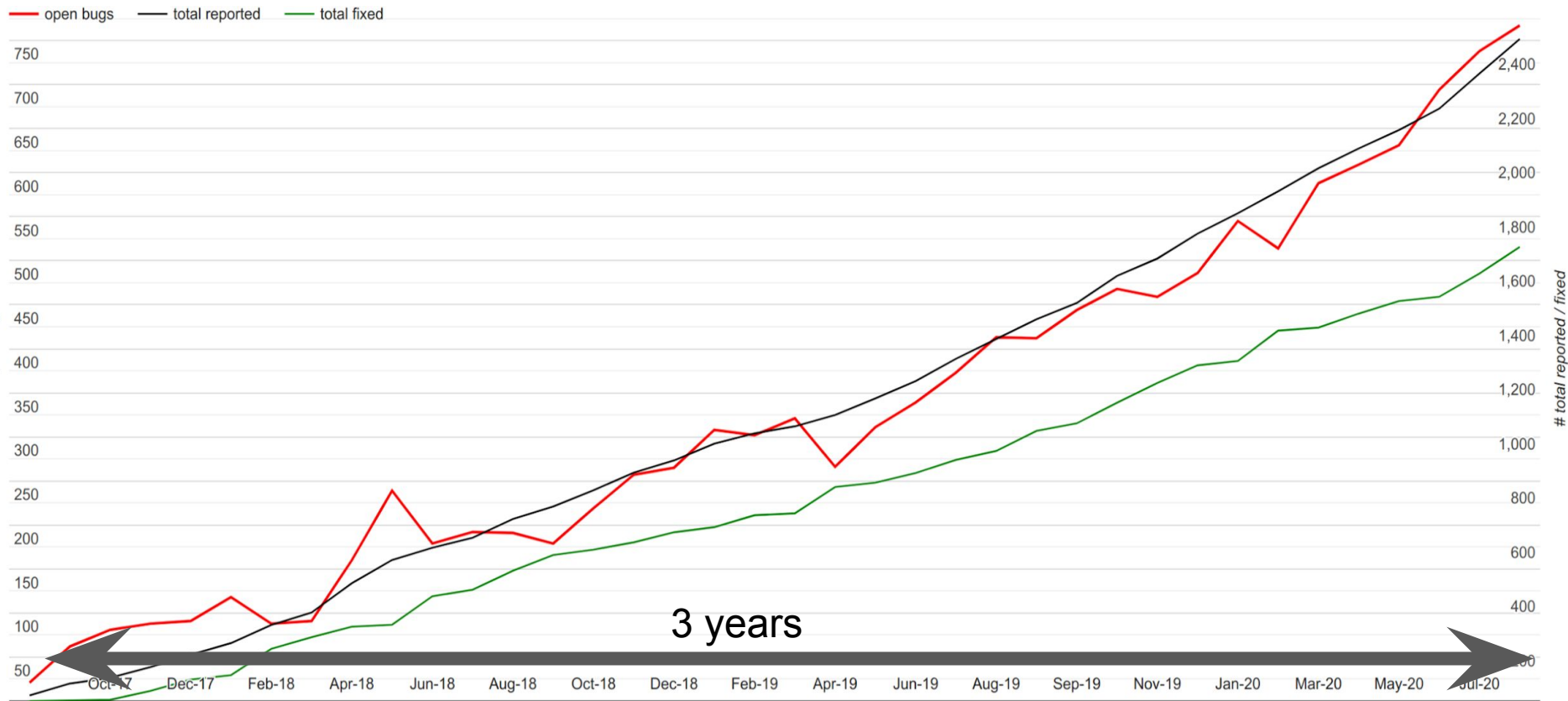
- Remind about unexisting fixing commit
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 - upstream: 80-120 days; linux-next: 40-60 days
 - based on crash rate
- Auto-upstreaming
 - syzkaller-upstream-moderation@googlegroups.com -> linux-kernel@vger.kernel.org
 - based on:
 - reproducer availability
 - number of crashes
 - bug type
 - tool (KCSAN/KMSAN)
 - edge-triggered -> level-triggered

stats and graphs

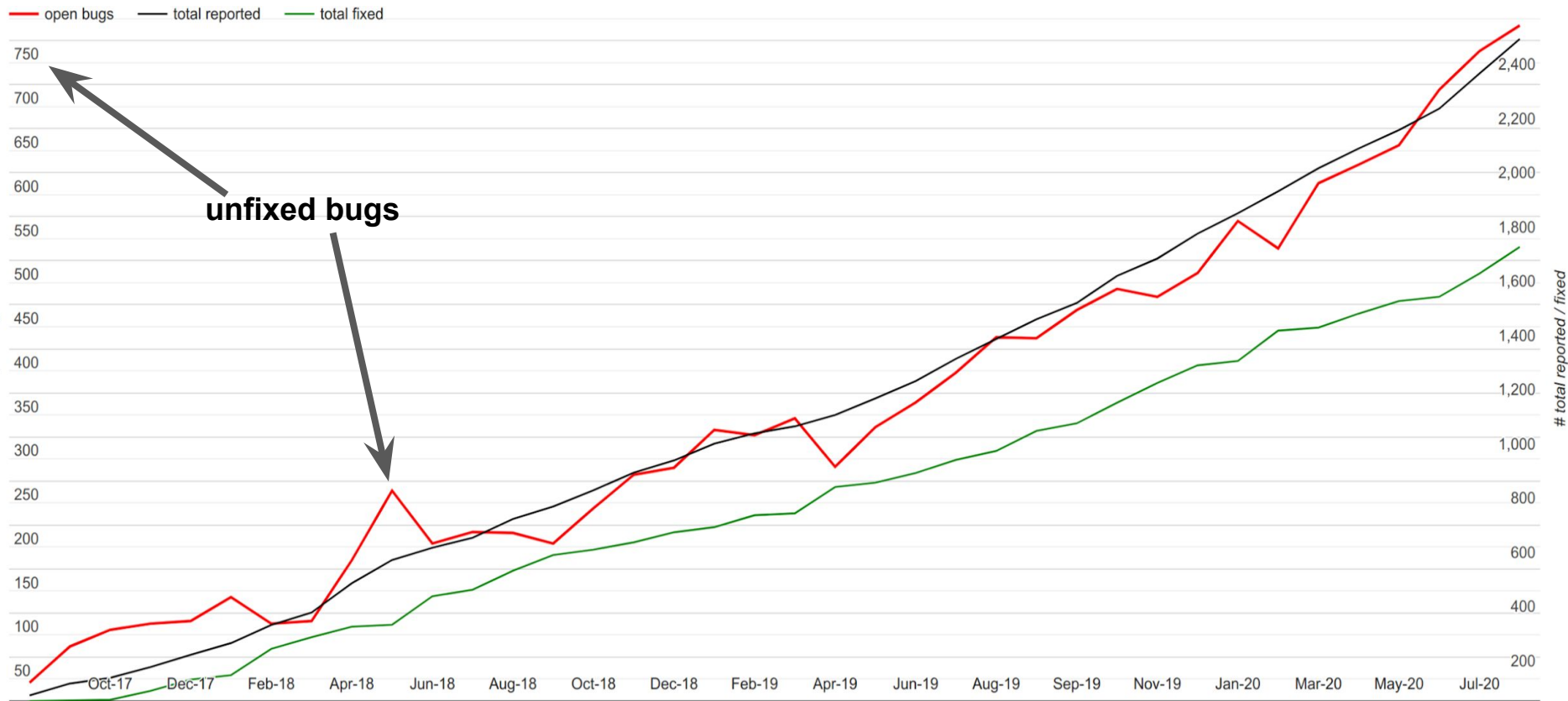
Upstream Bug Stats



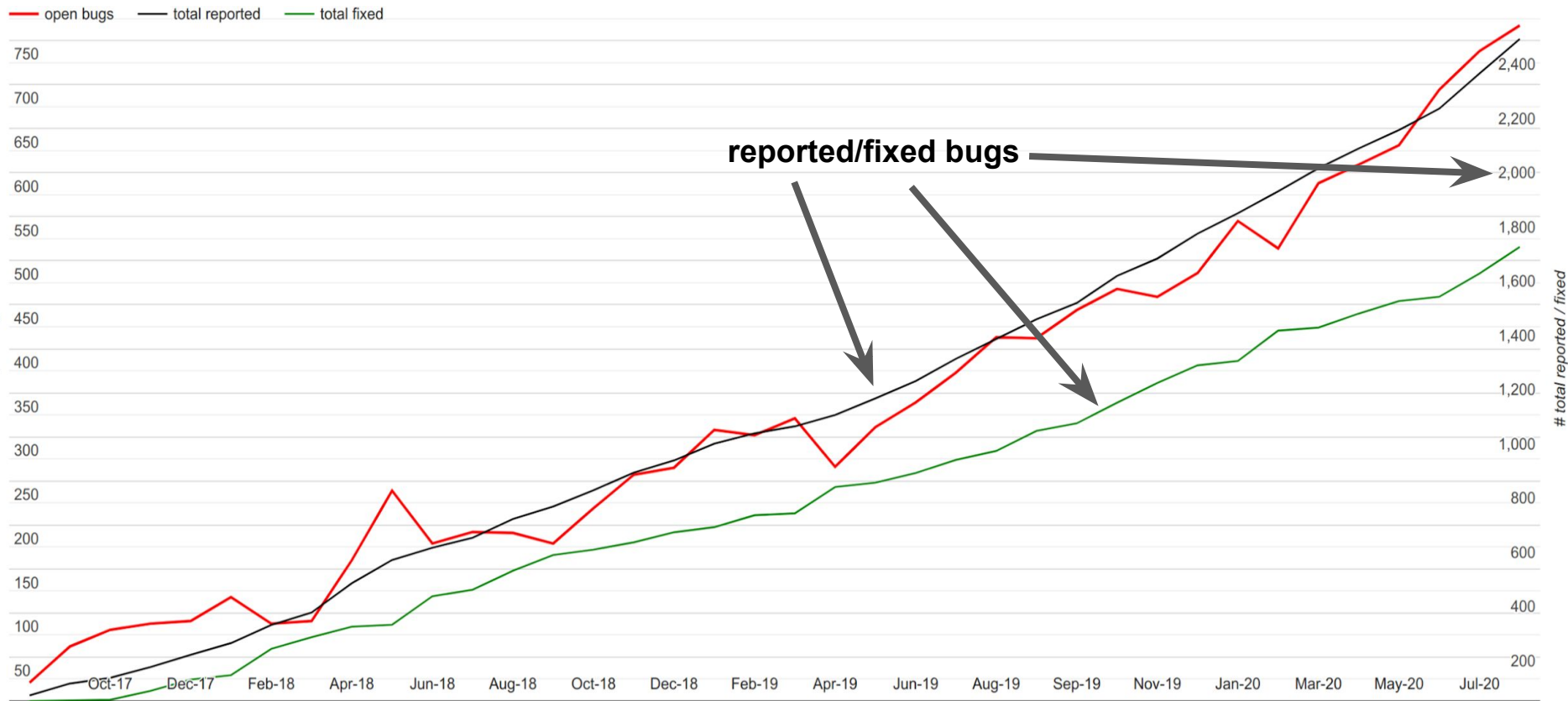
Upstream Bug Stats



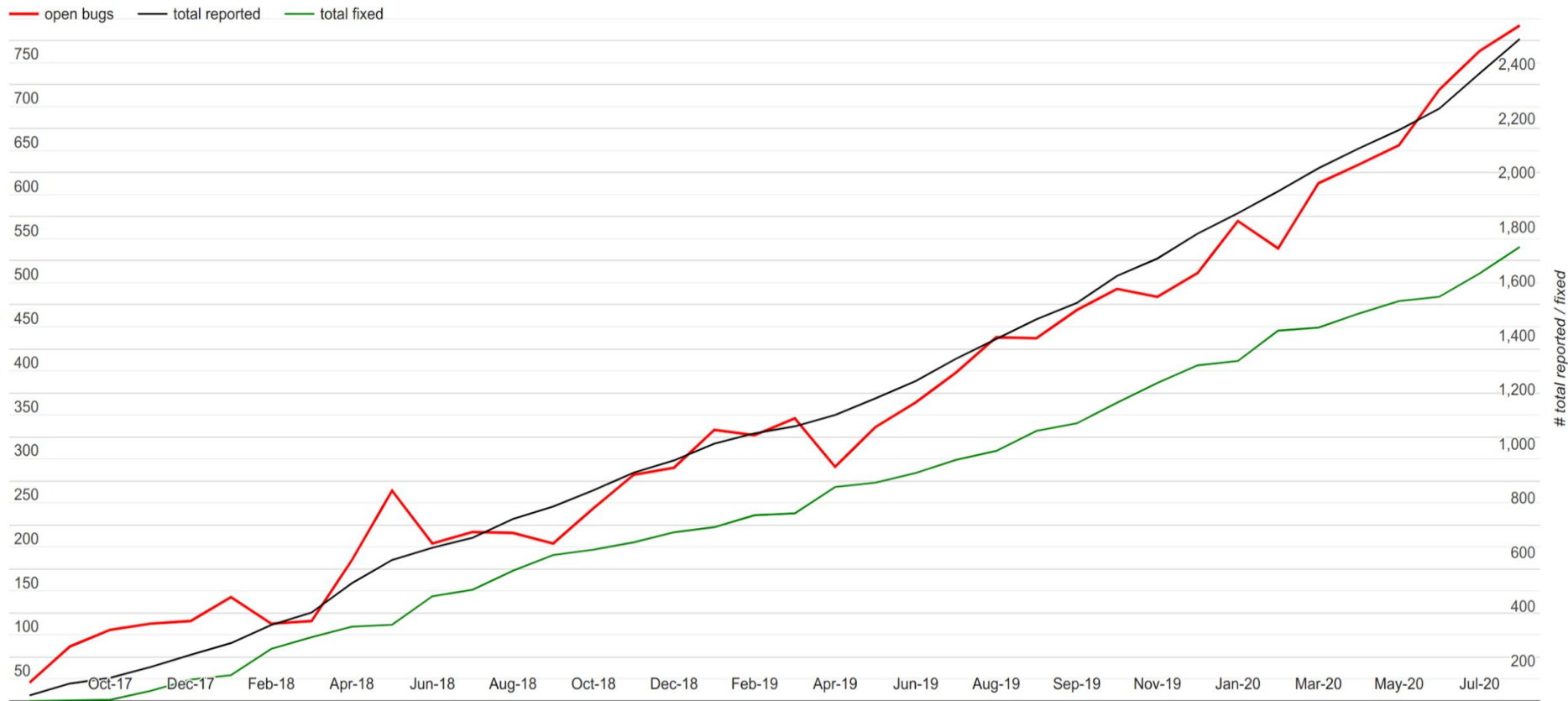
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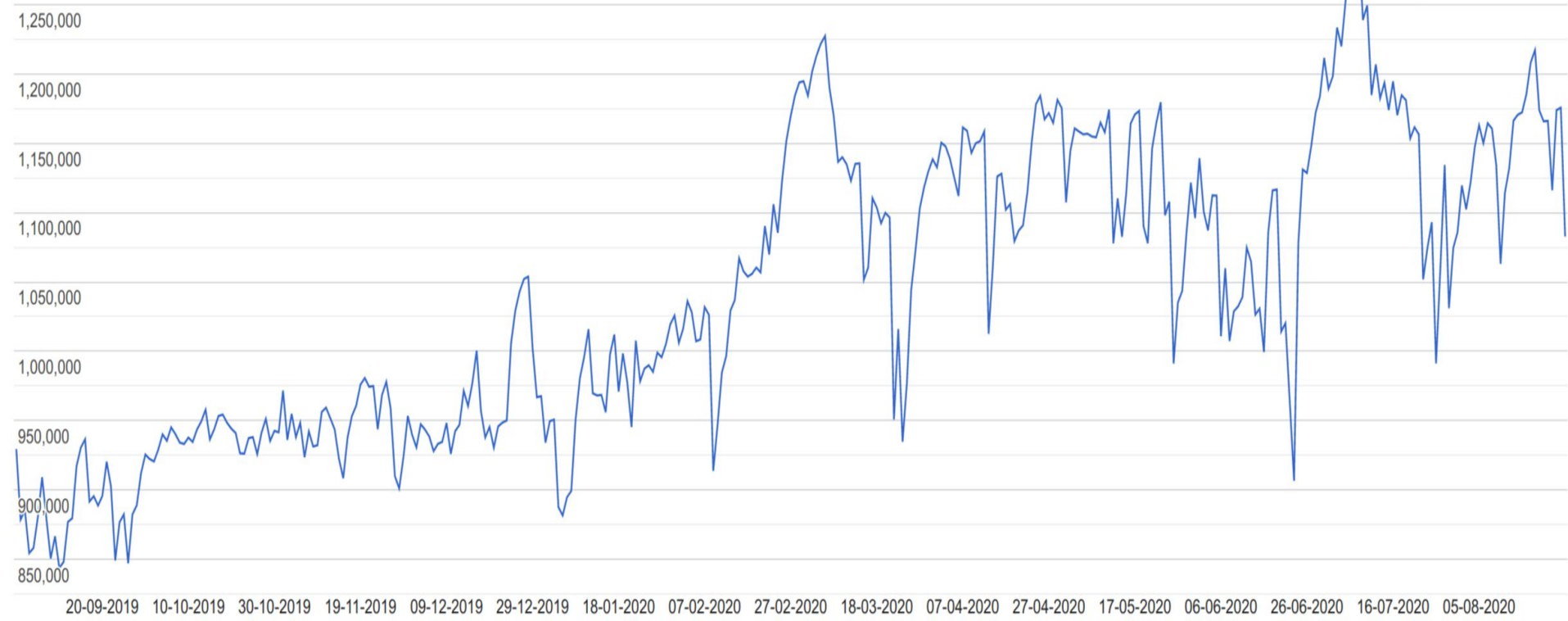


Upstream Bug Stats



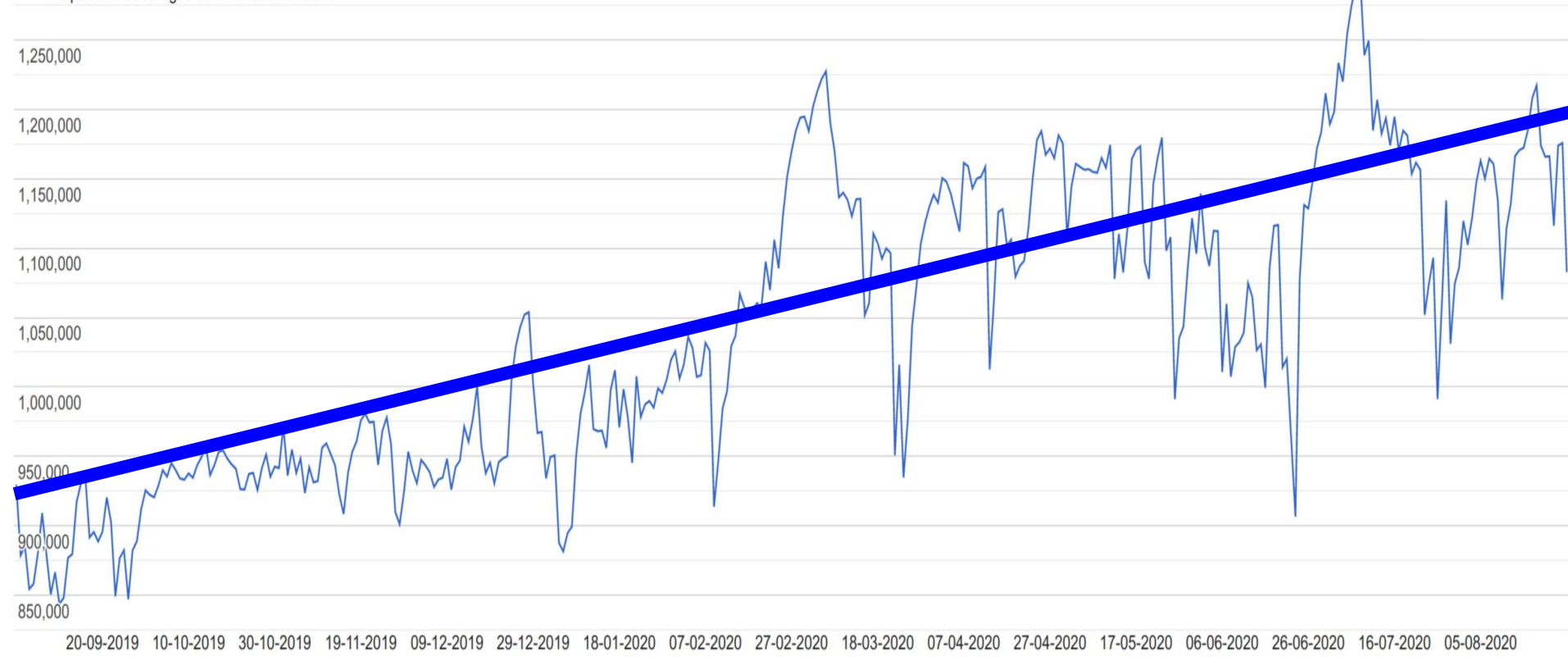
Upstream kernel coverage

ci-upstream-kasan-gce-selinux-root-MaxCover



Upstream kernel coverage

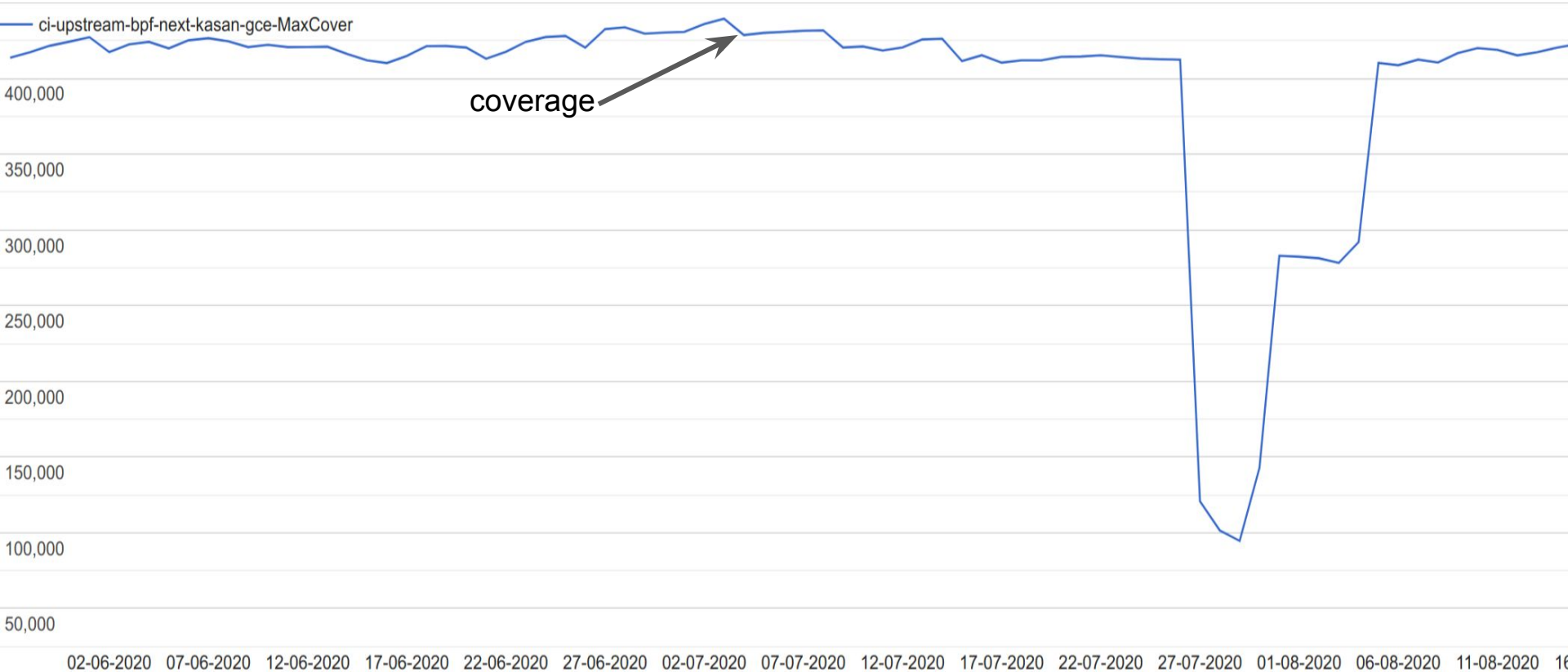
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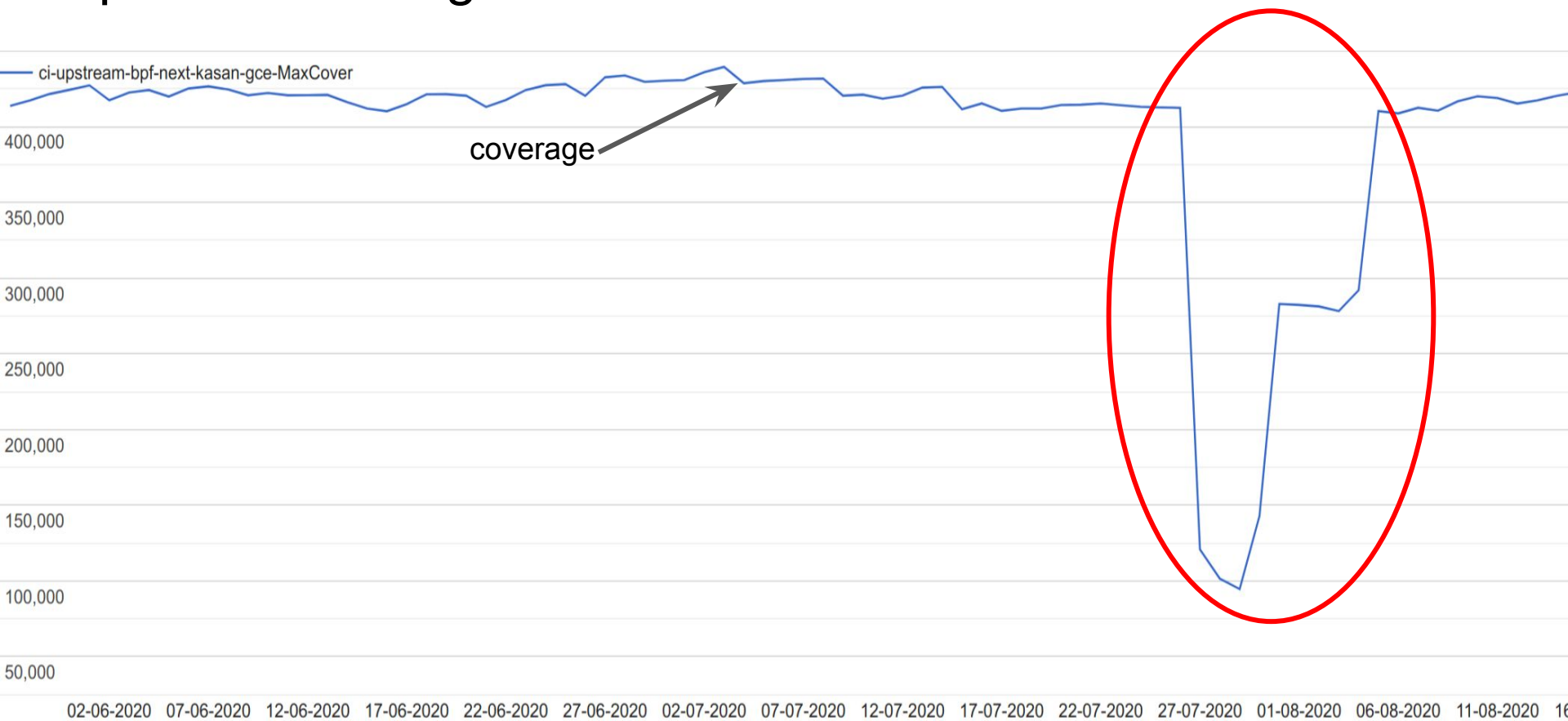
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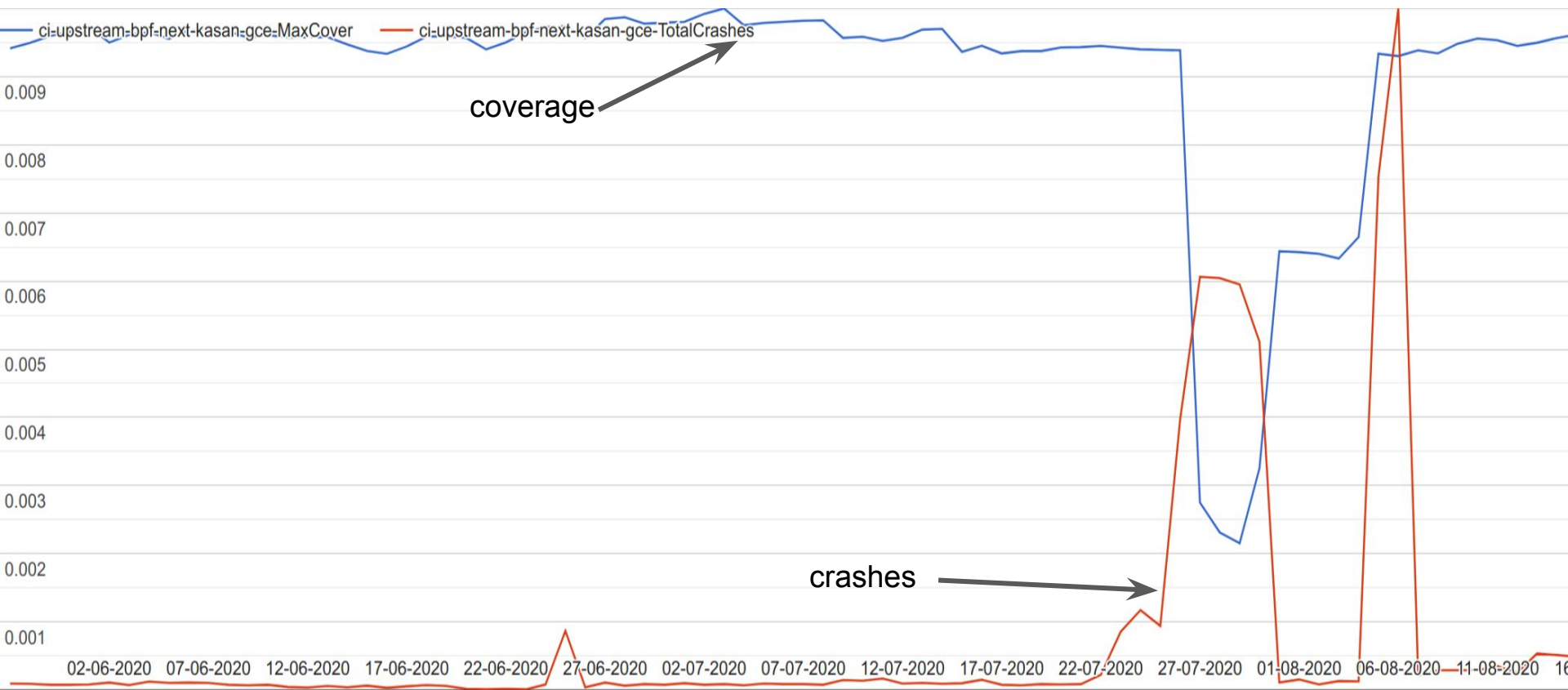
bpf-next coverage



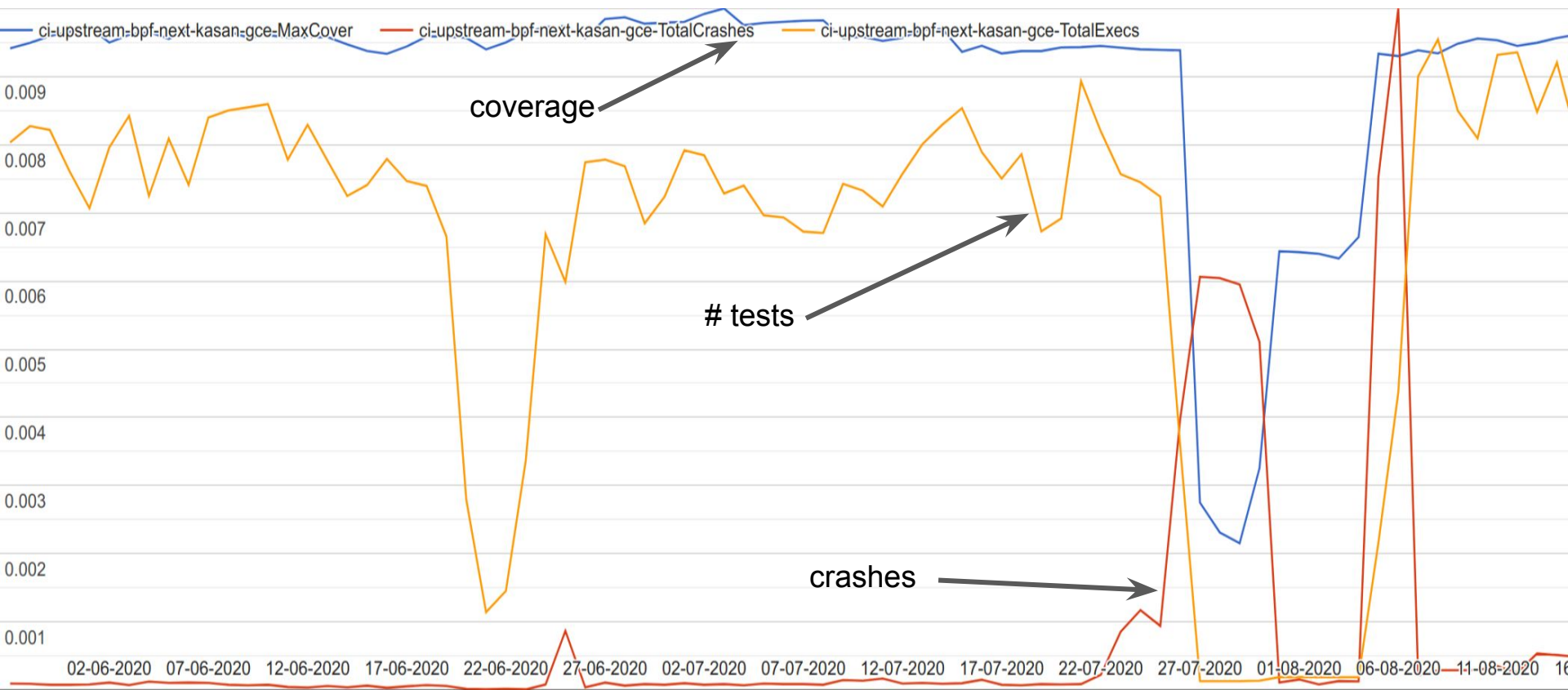
bpf-next coverage



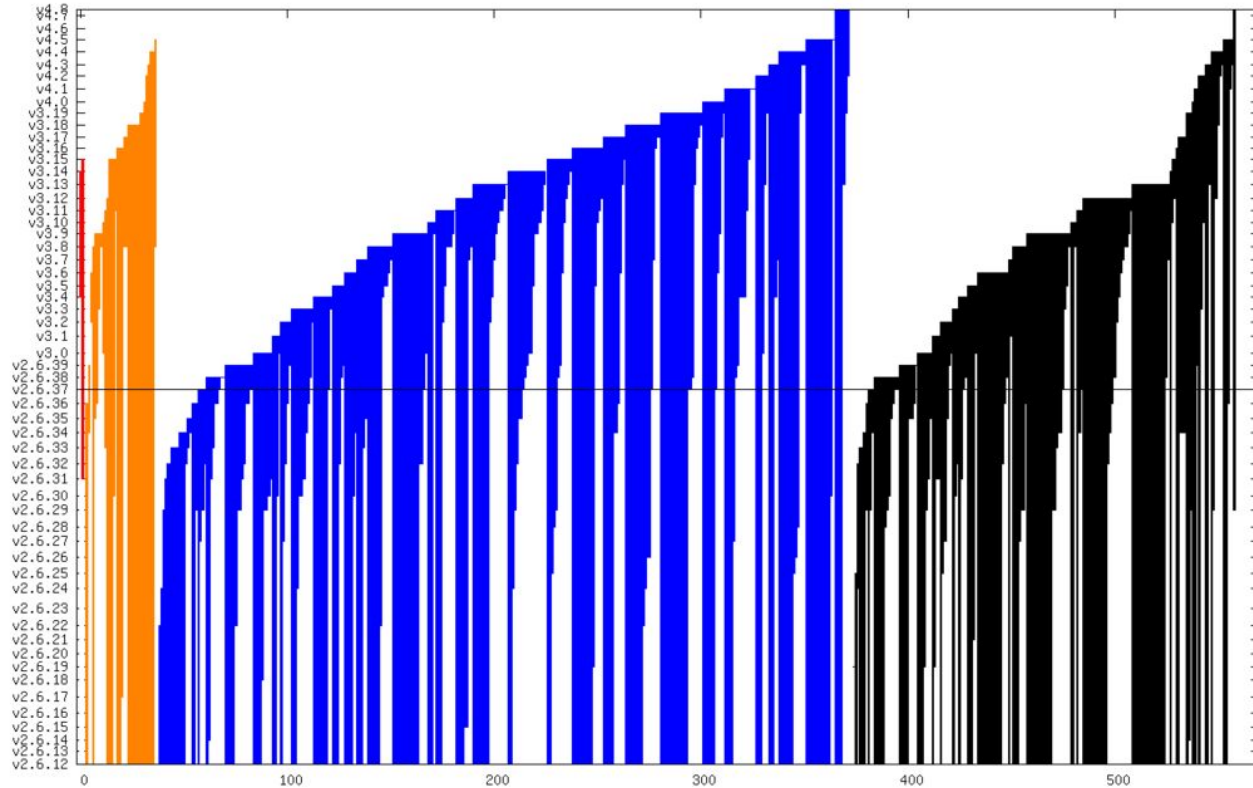
bpf-next coverage



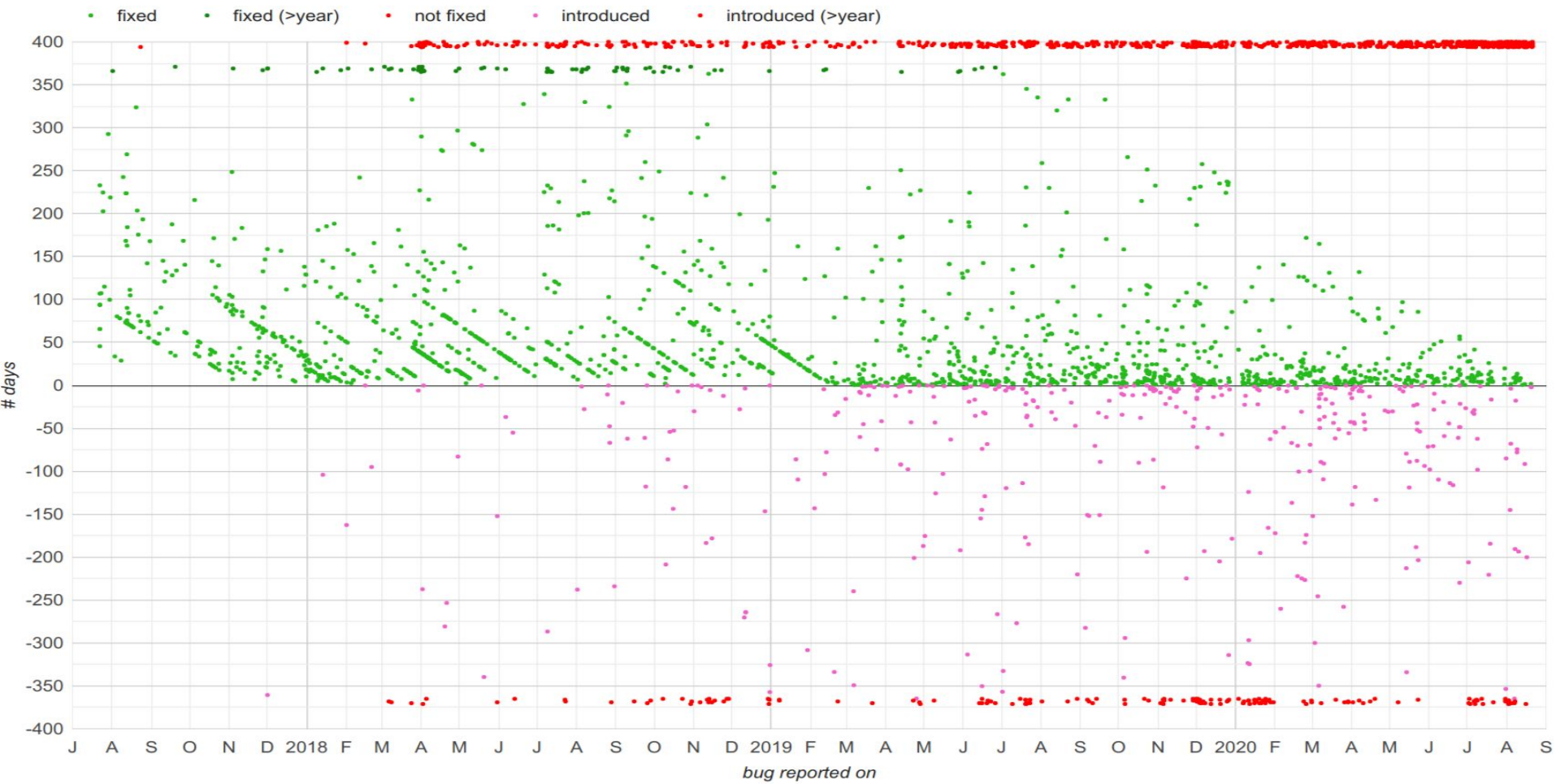
bpf-next coverage



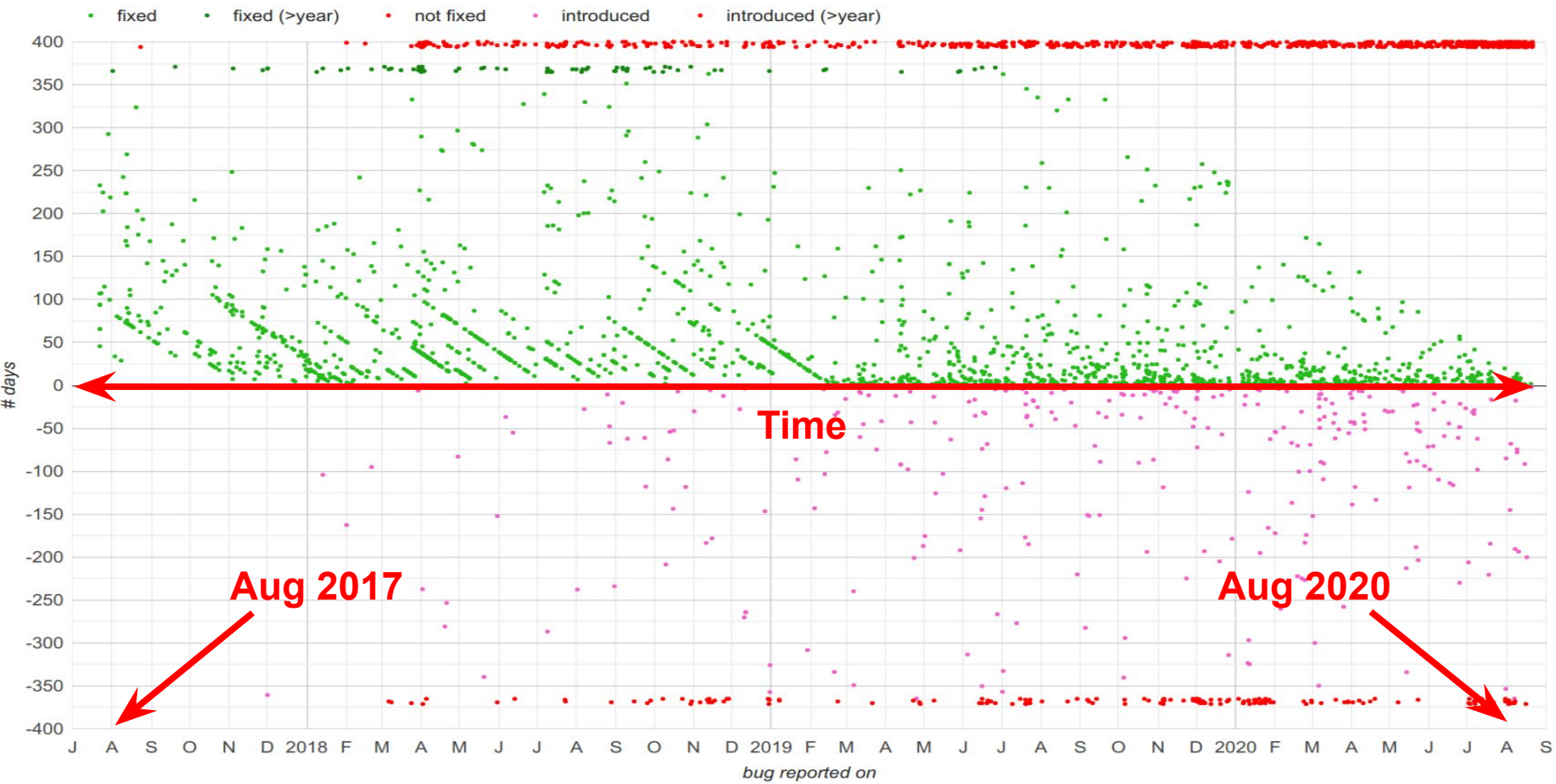
Security bug lifetime (© Kees Cook)



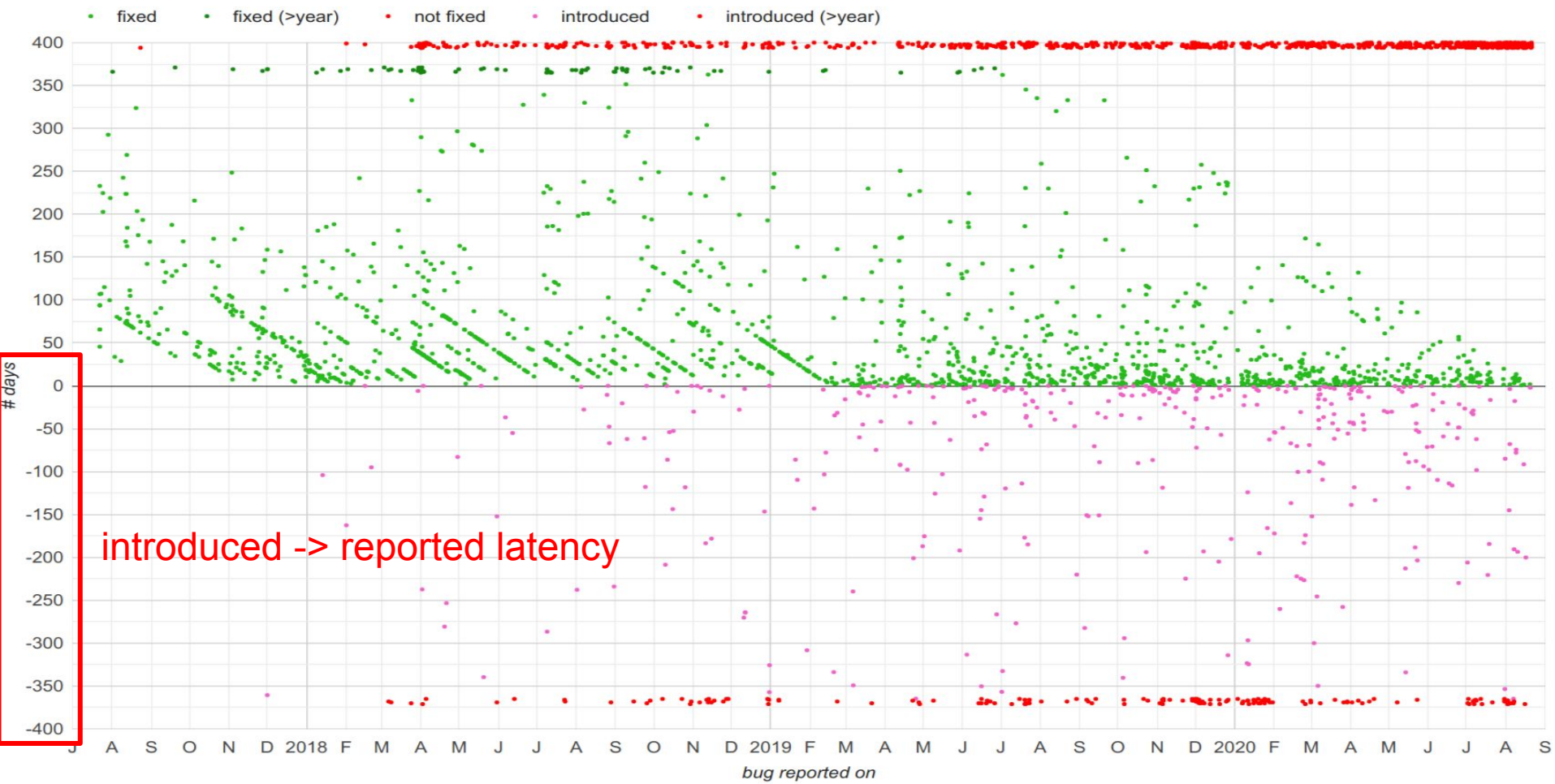
Bug Lifetimes



Bug Lifetimes



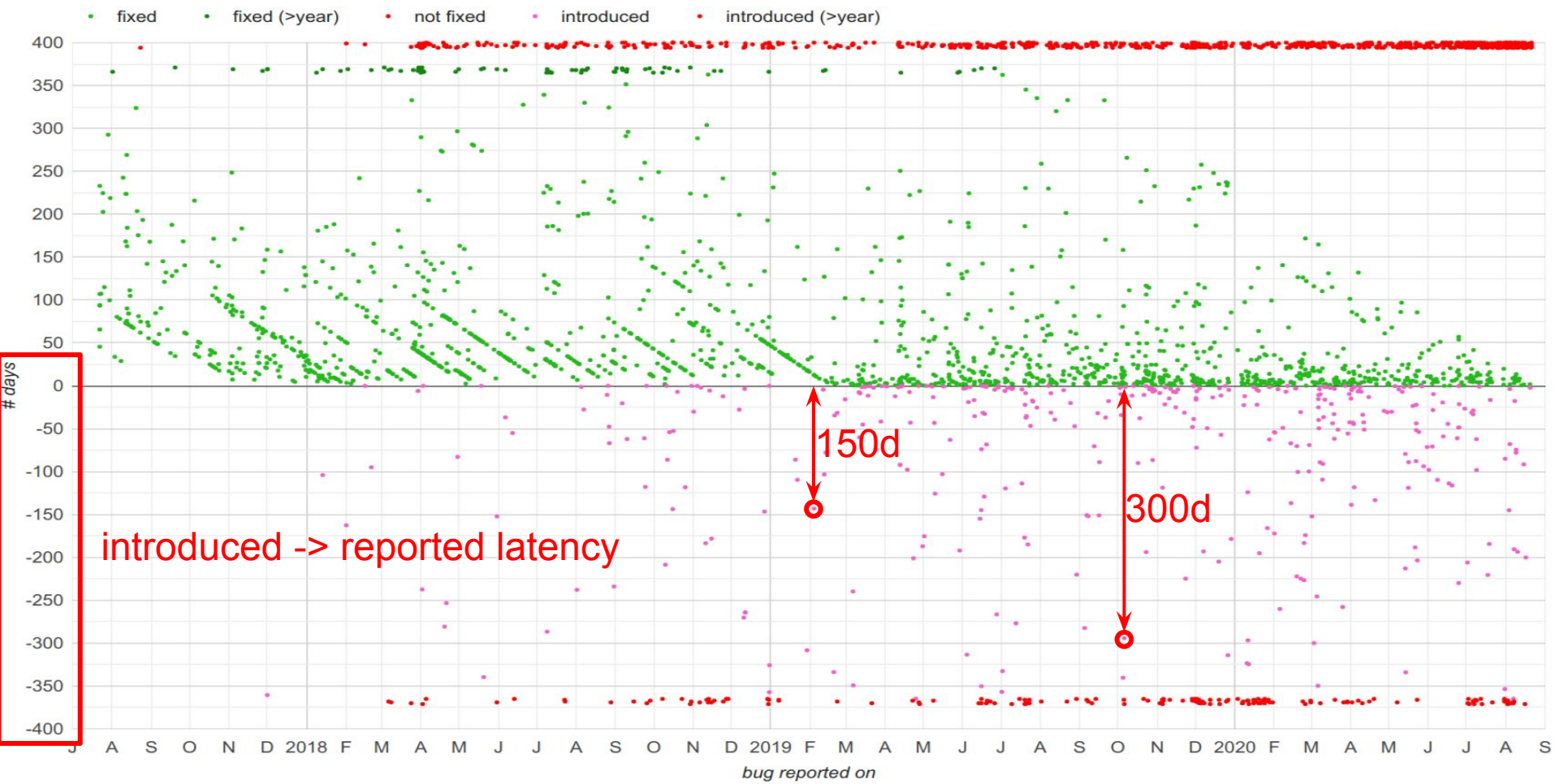
Bug Lifetimes



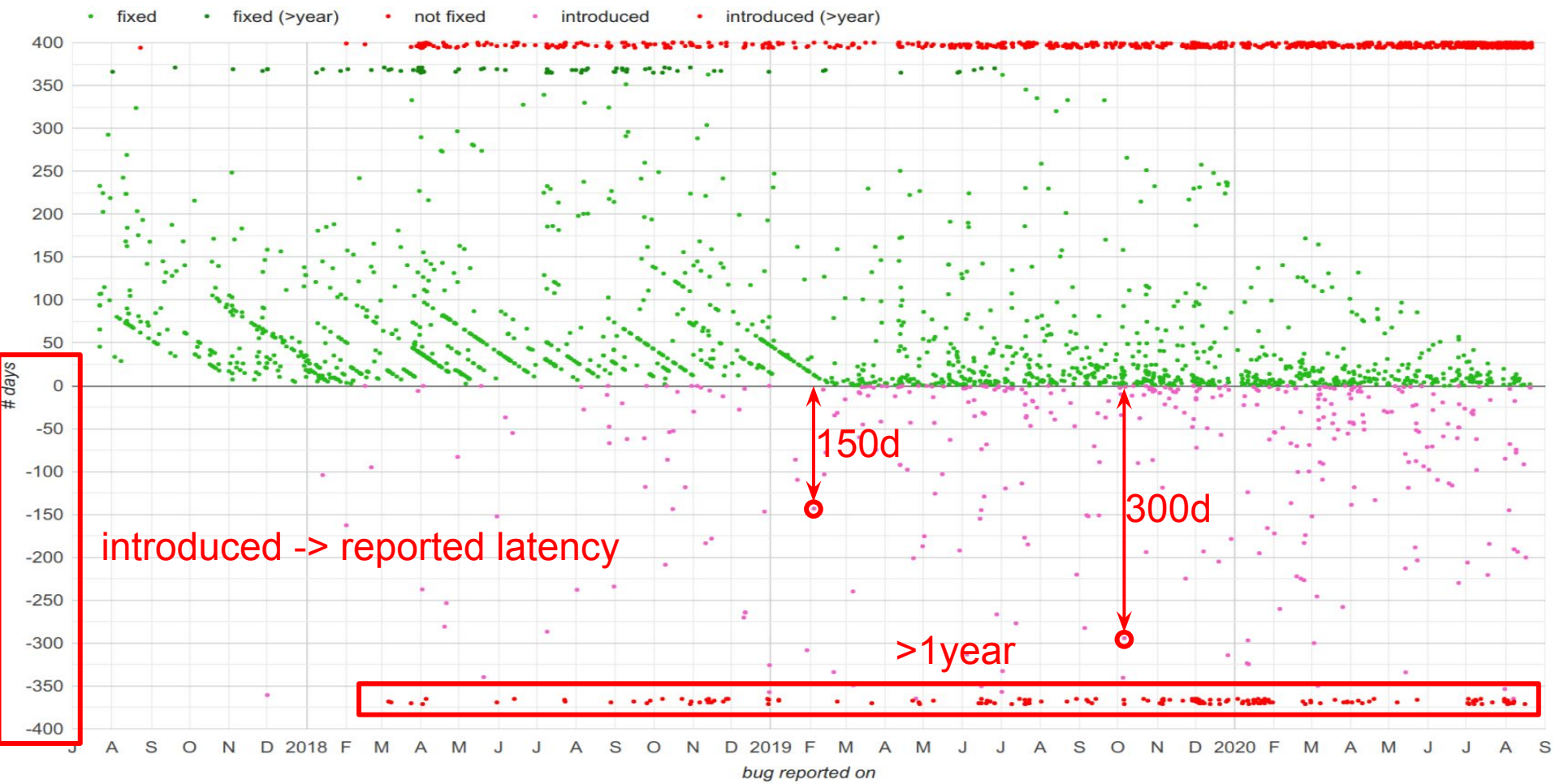
Bug Lifetimes



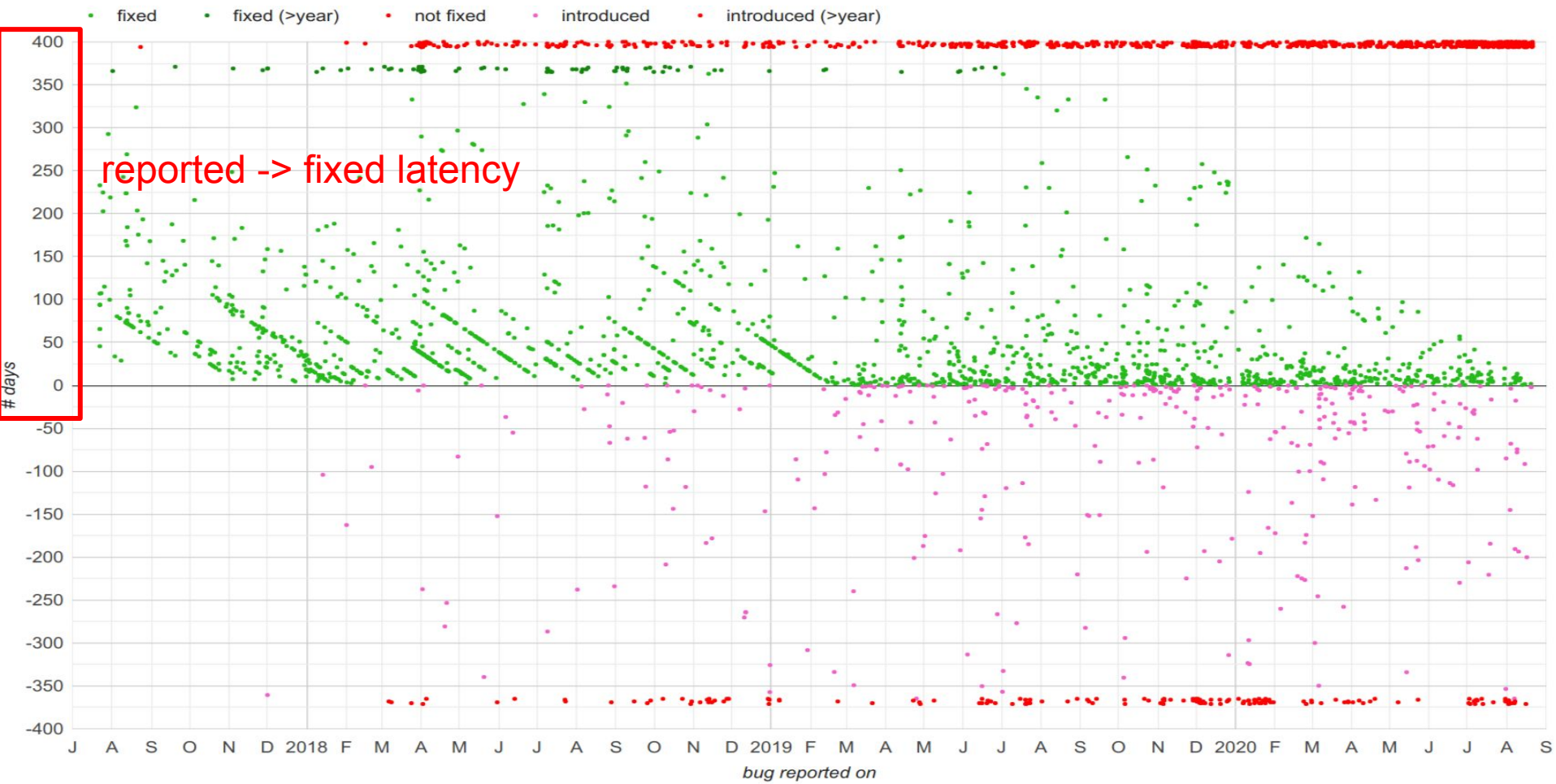
Bug Lifetimes



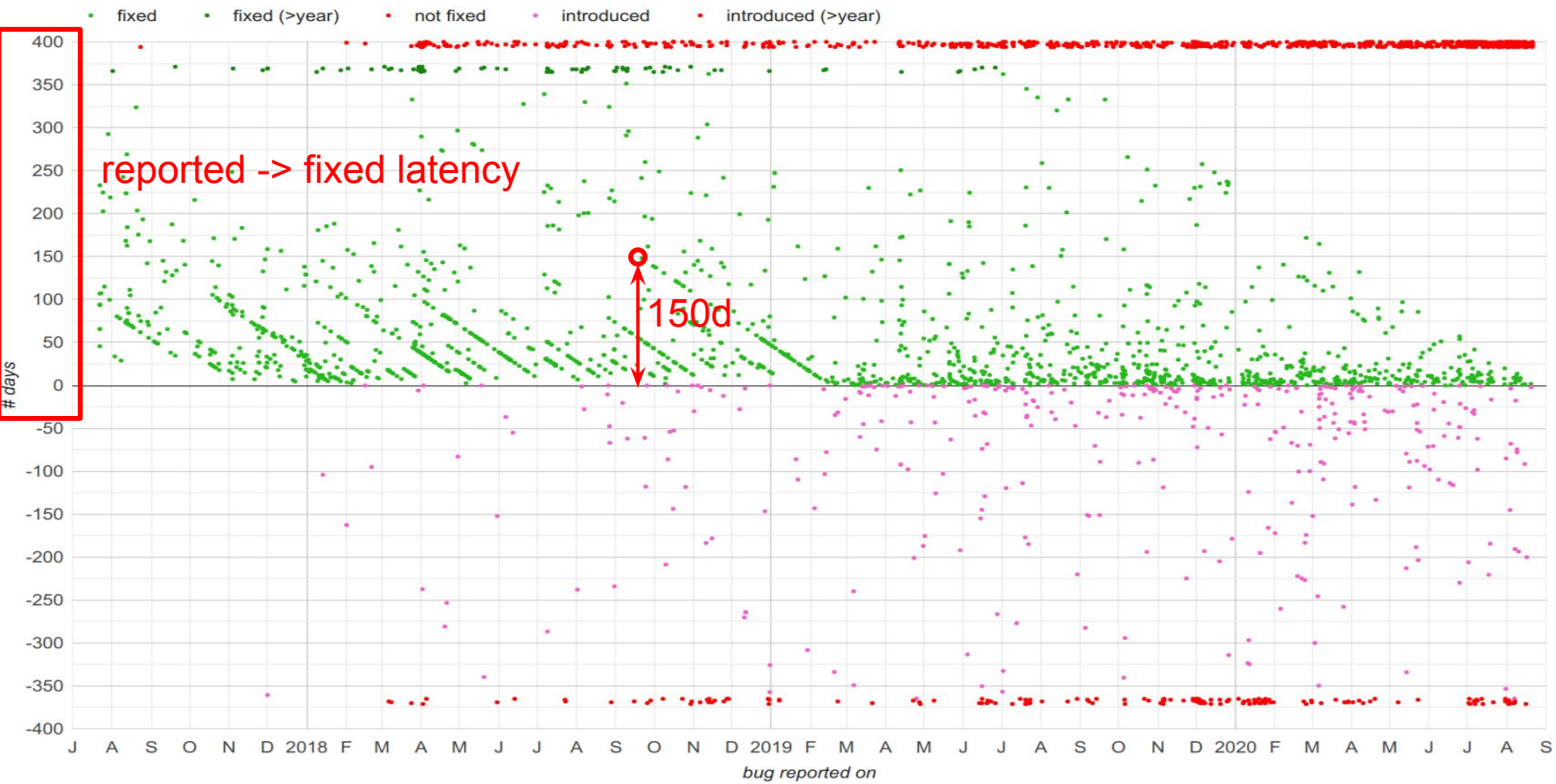
Bug Lifetimes



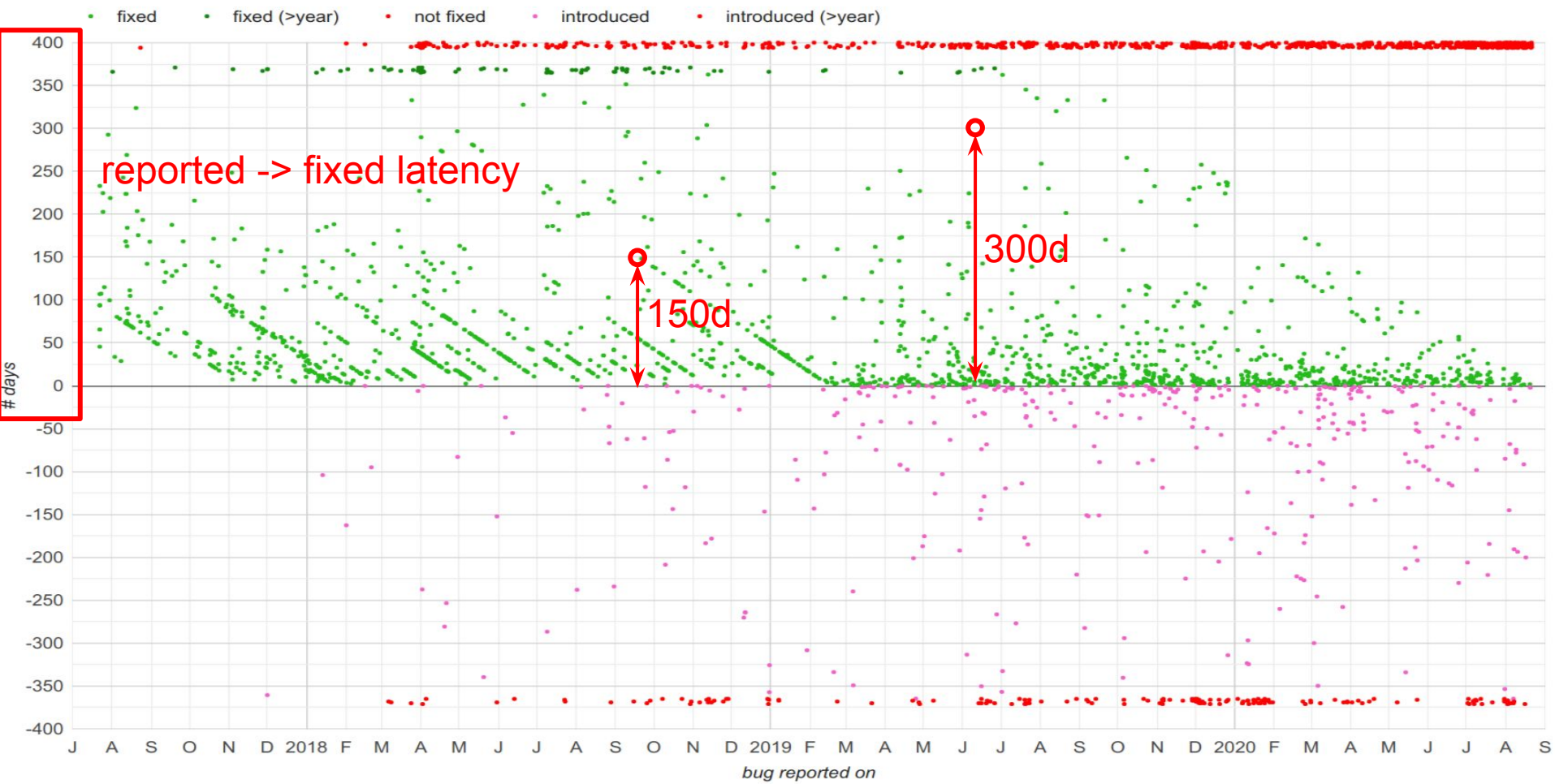
Bug Lifetimes



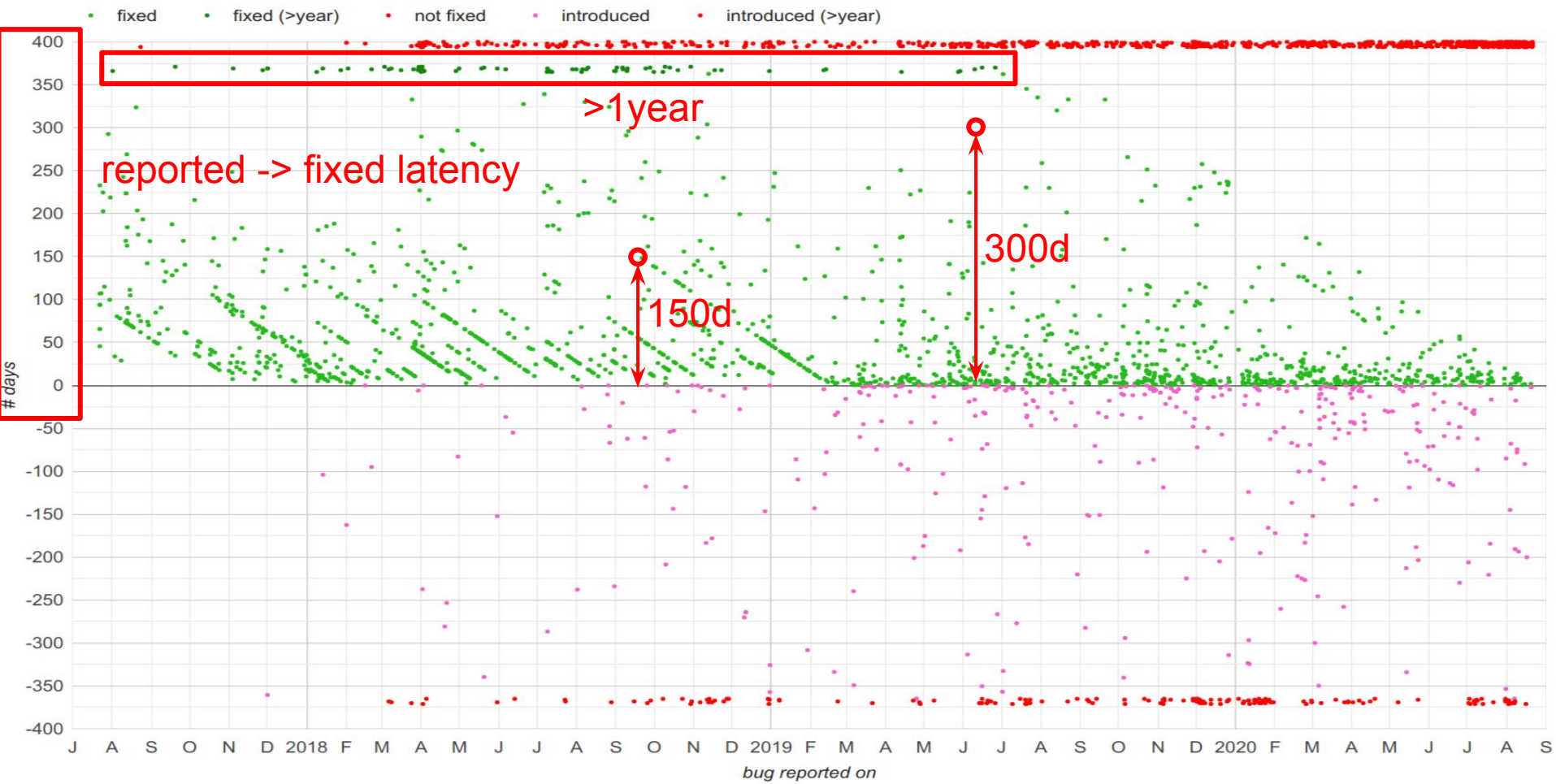
Bug Lifetimes



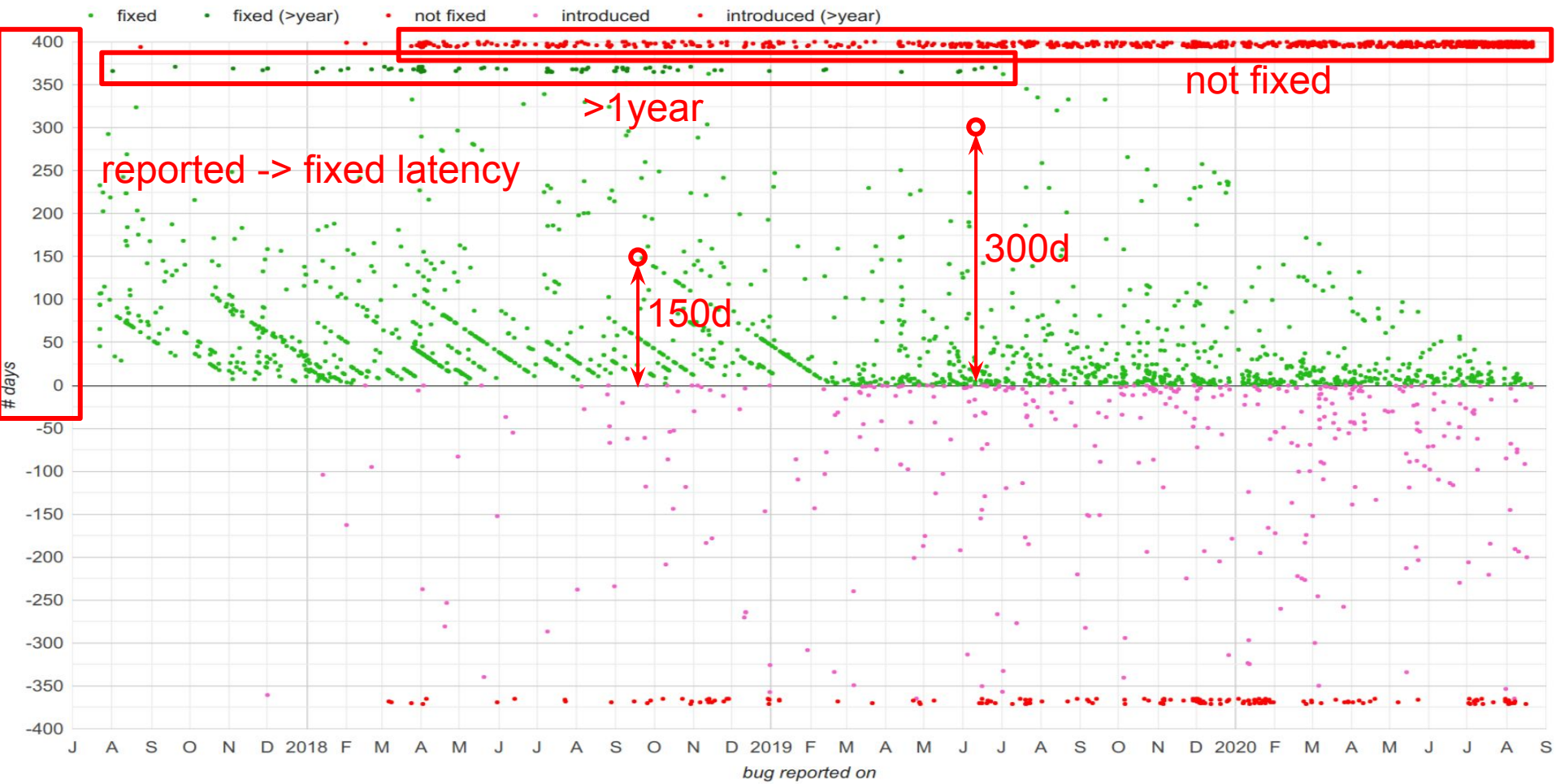
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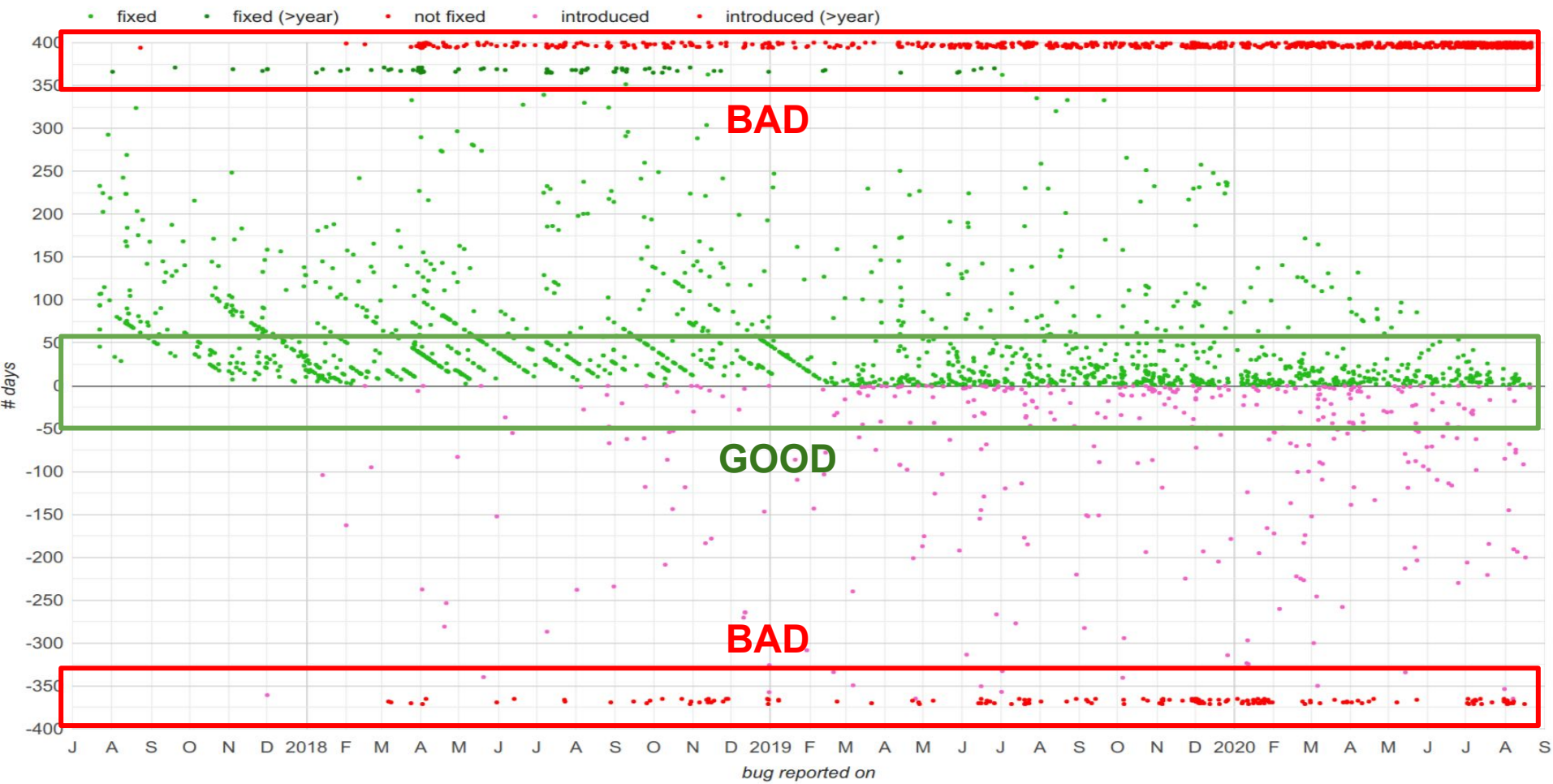
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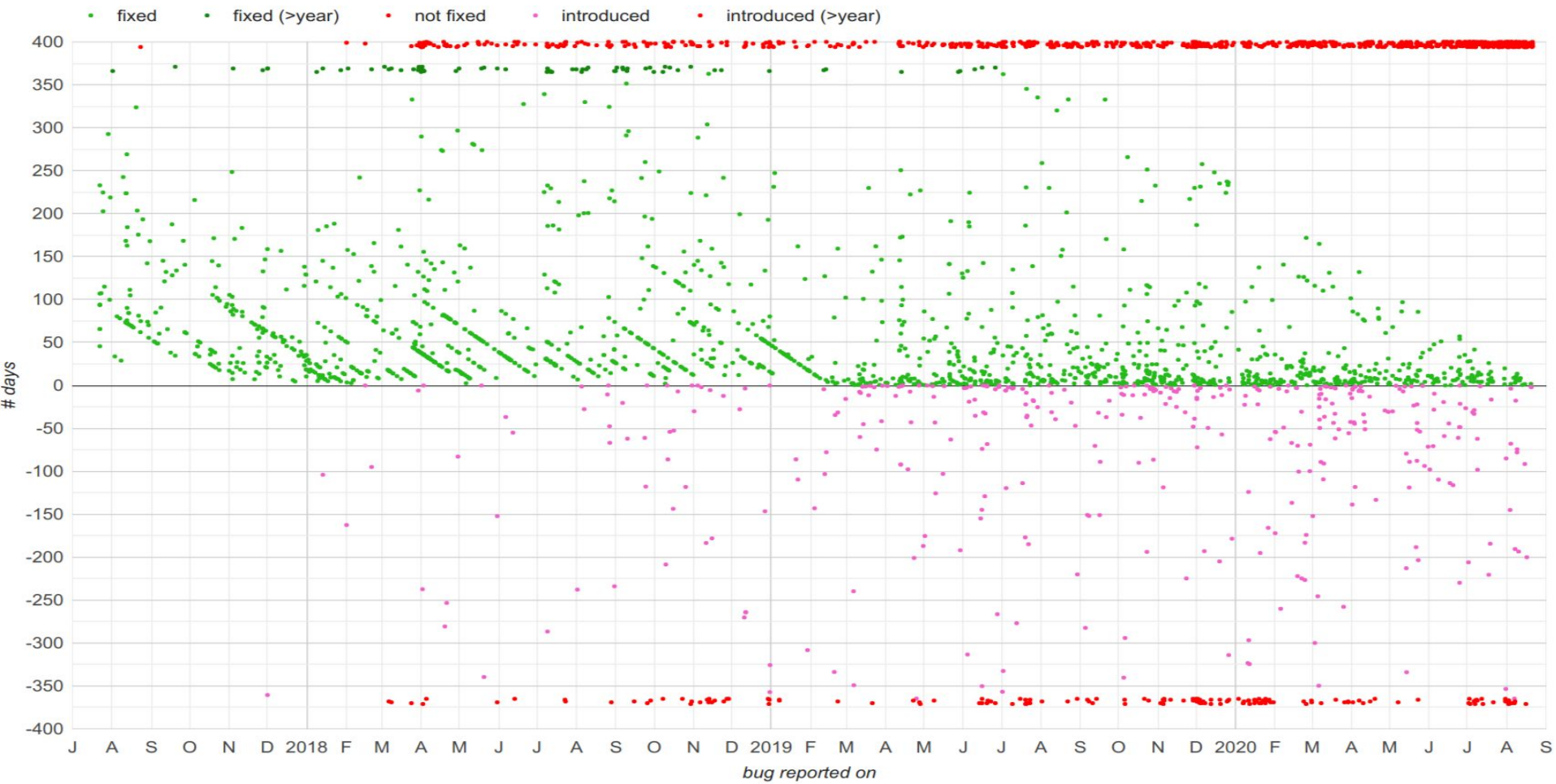
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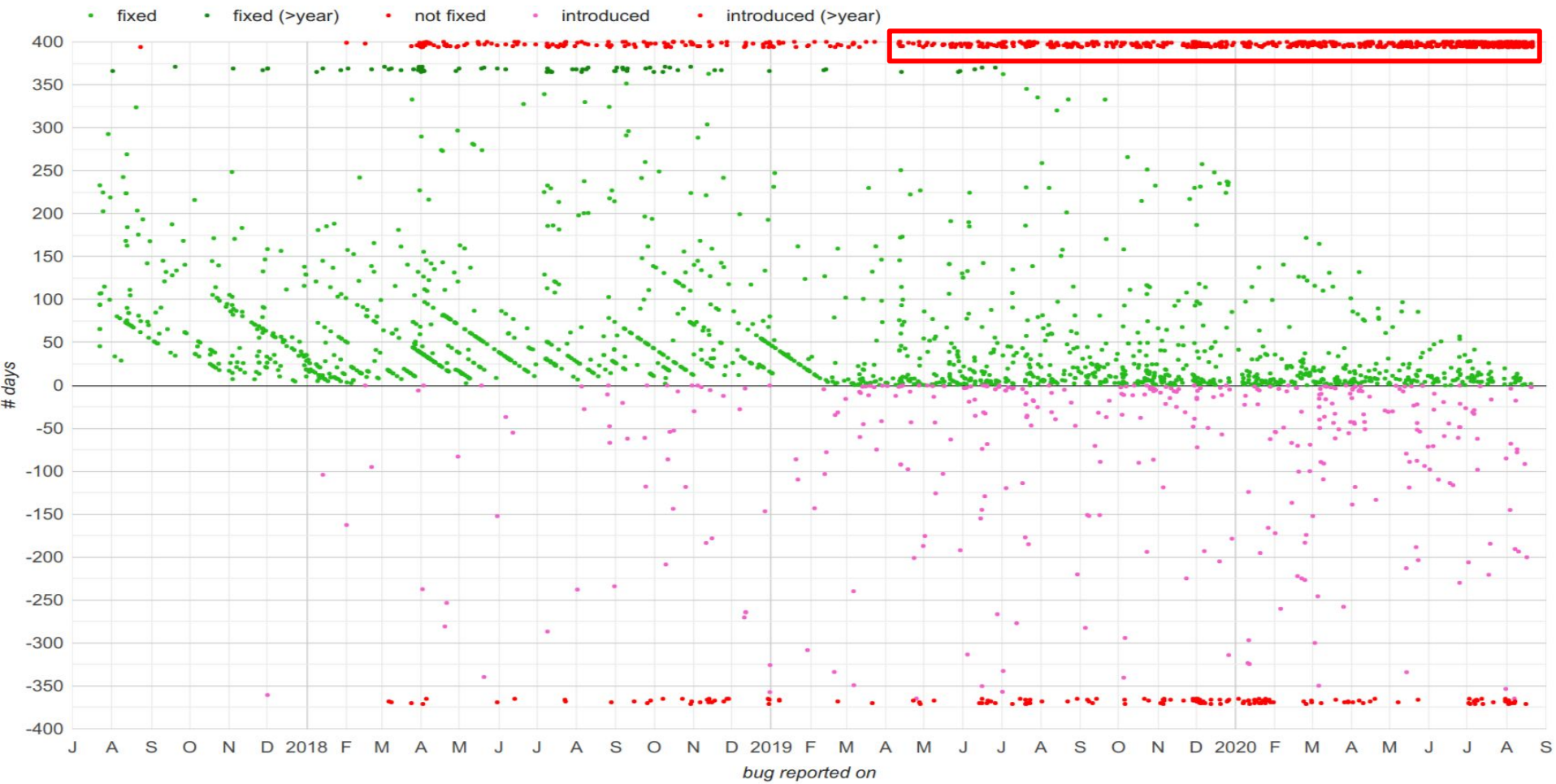
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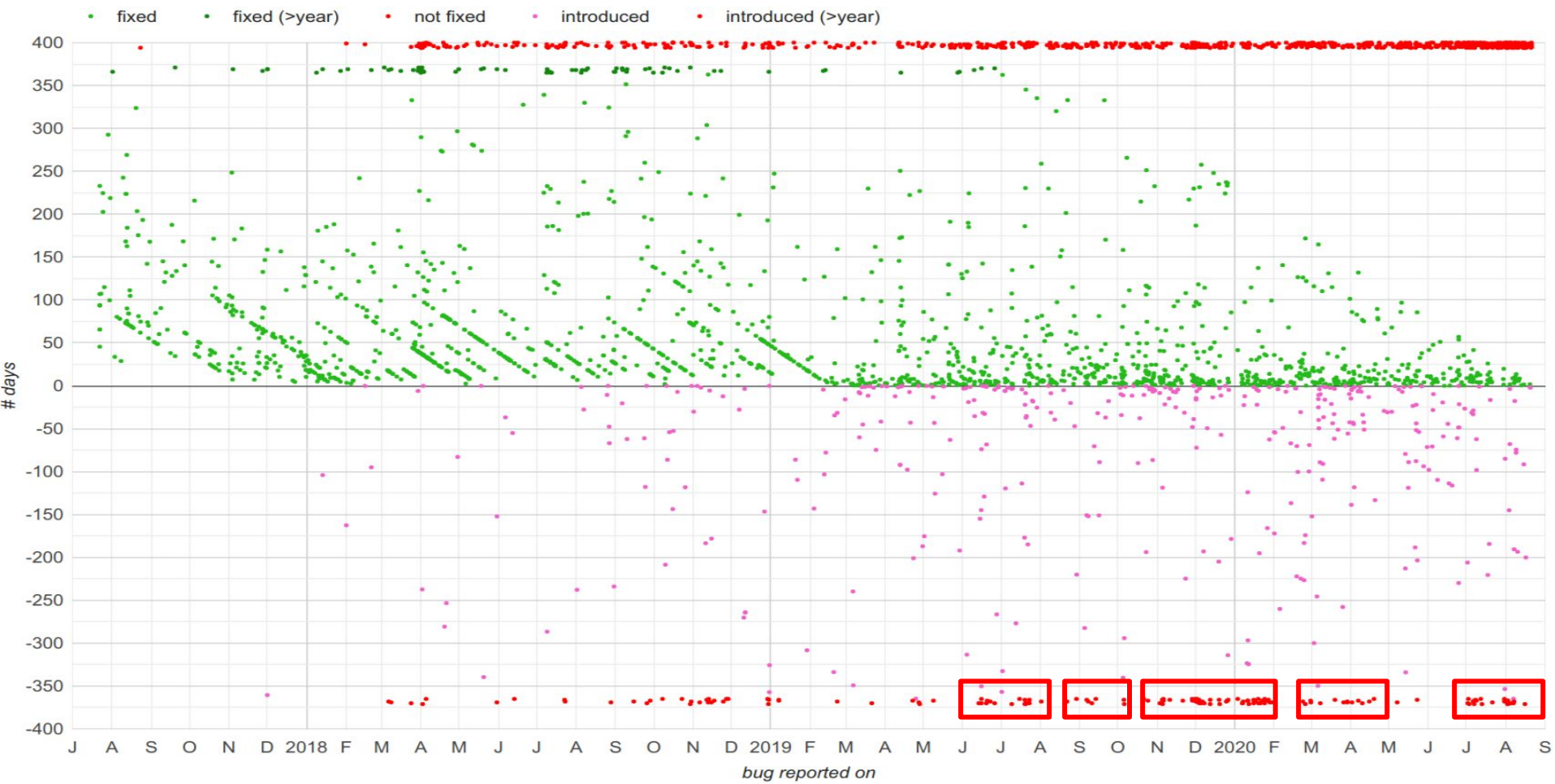
Bug Lifetimes



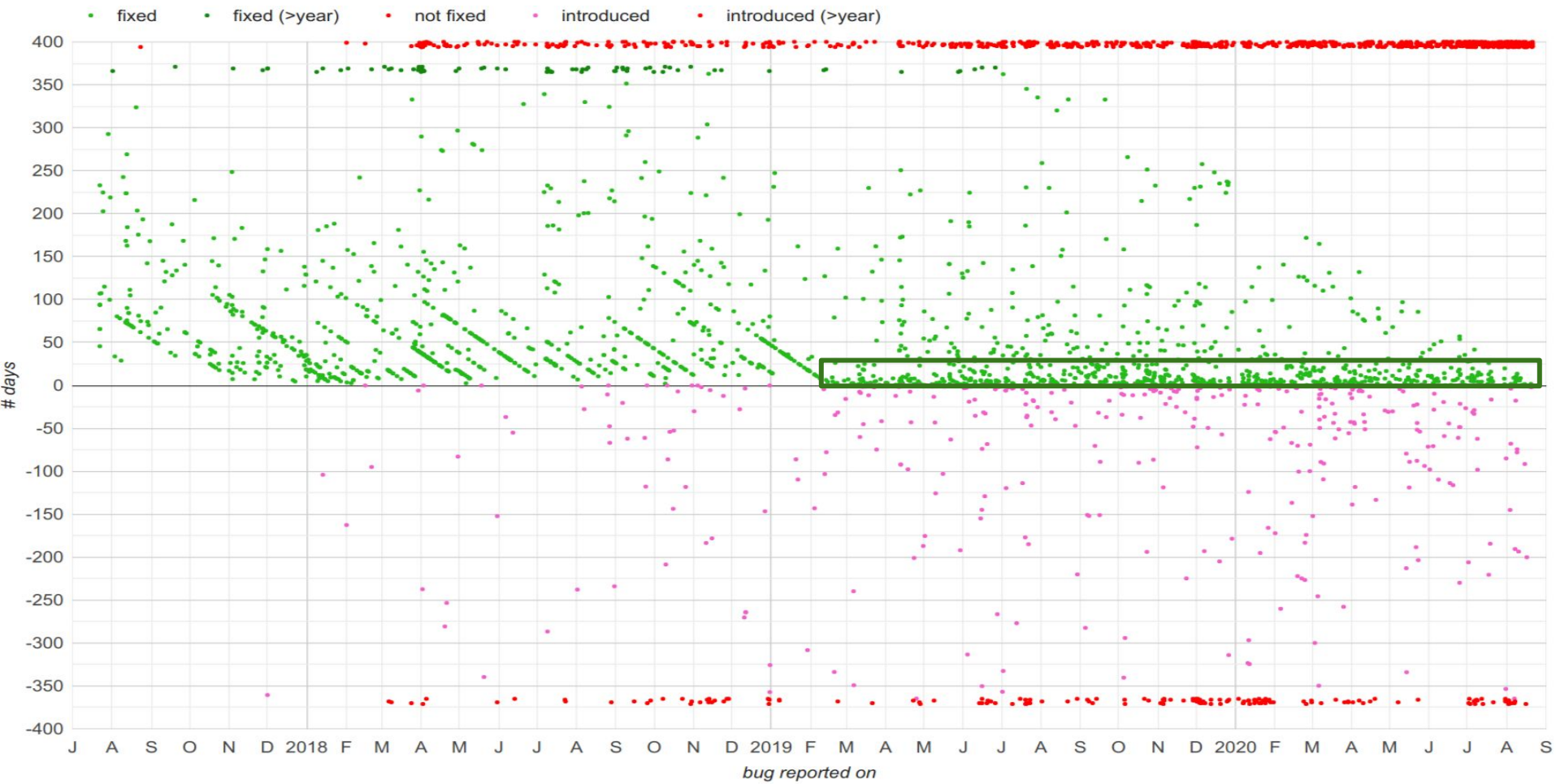
Bug Lifetimes



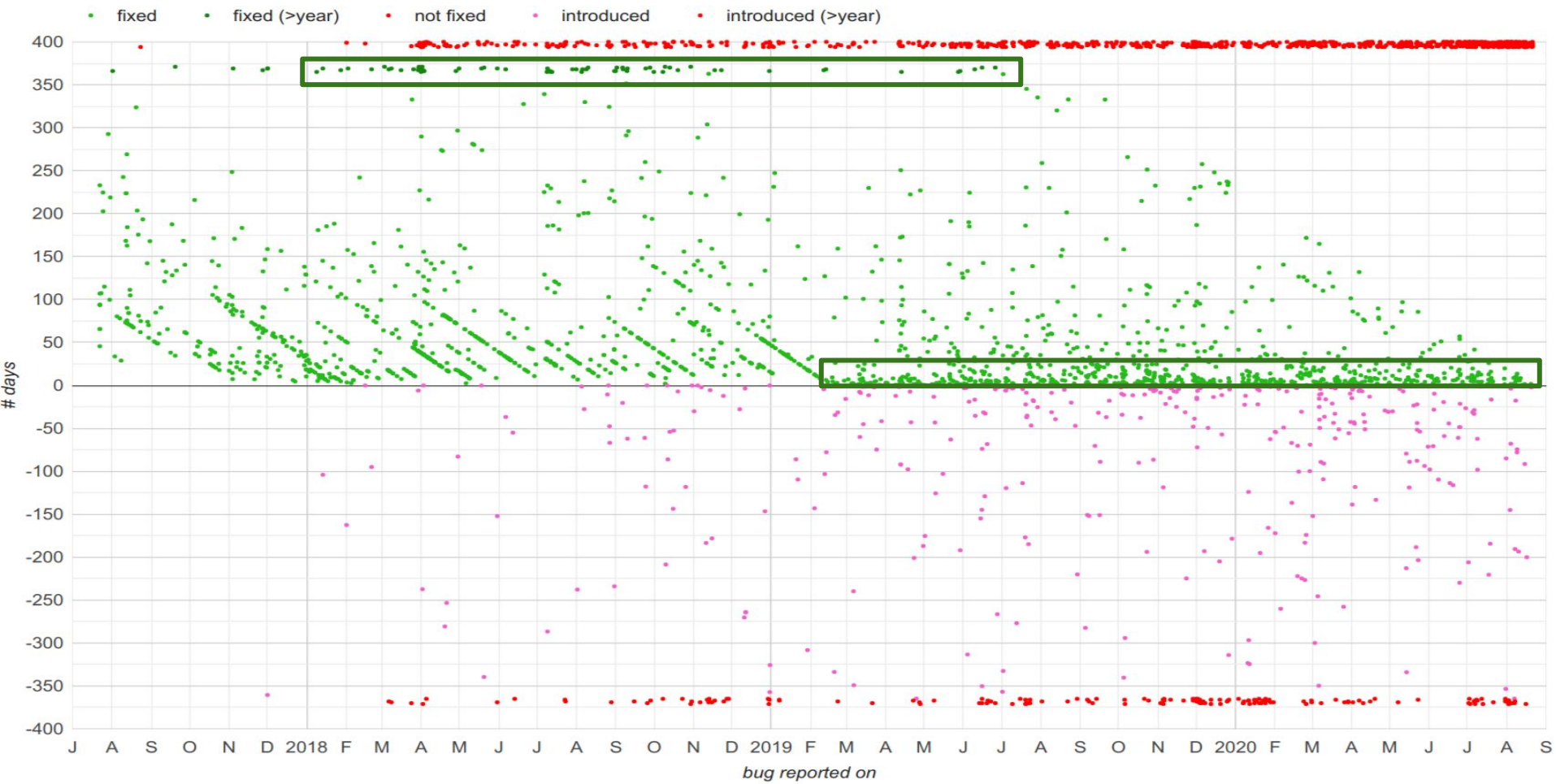
Bug Lifetimes



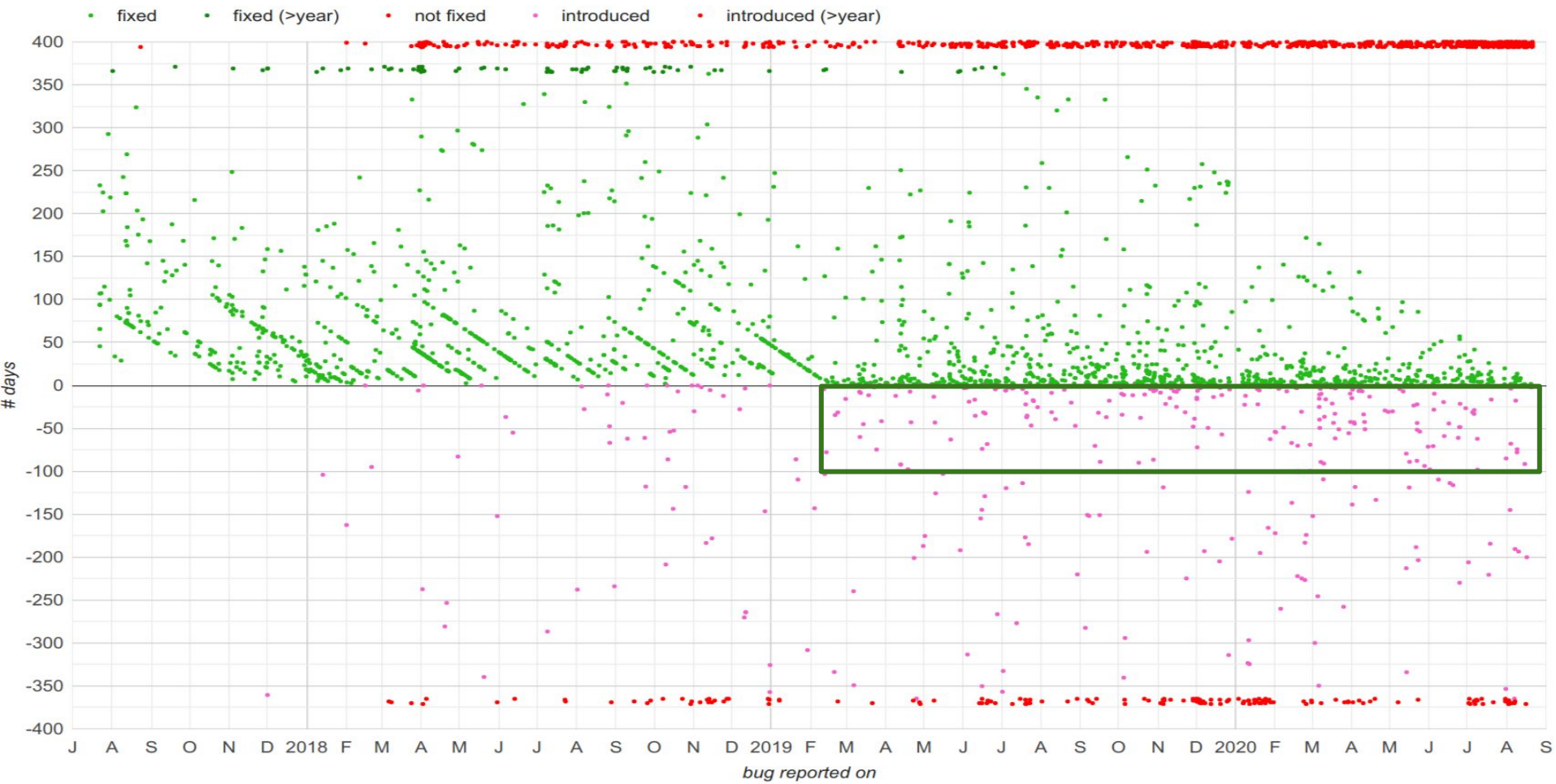
Bug Lifetimes



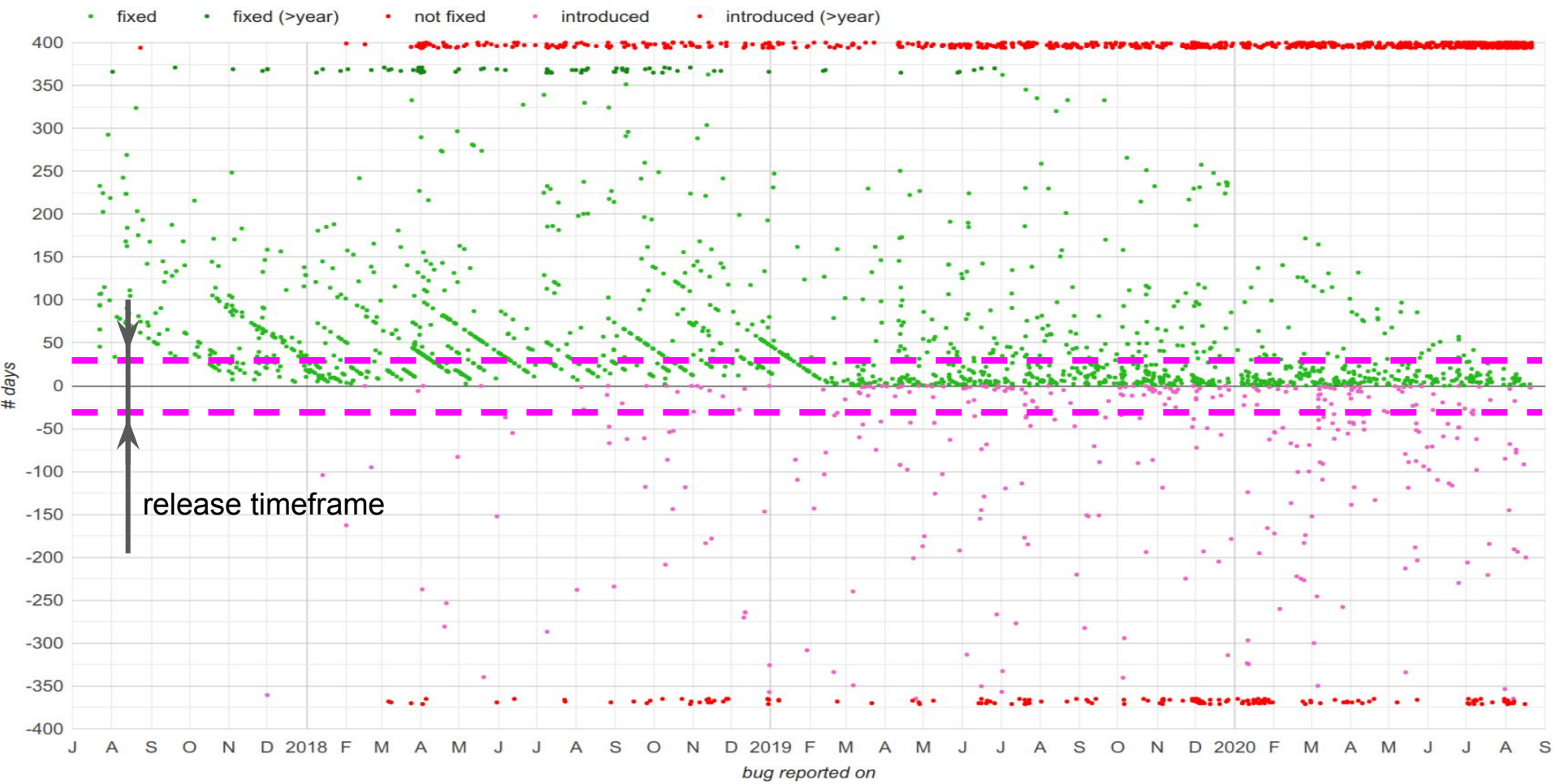
Bug Lifetimes



Bug Lifetimes



Bug Lifetimes



Thank you!

Q&A

`syzkaller@googlegroups.com`
`kasan-dev@googlegroups.com`

Dmitry Vyukov (`dvyukov@`)

LTS

	Open	"Fixed"	"Obsolete"
4.19	368	140	189
4.14	506	100	22
4.9	WARNING in cpumask_check		
4.4	WARNING in batadv_tvlv_container_remove		